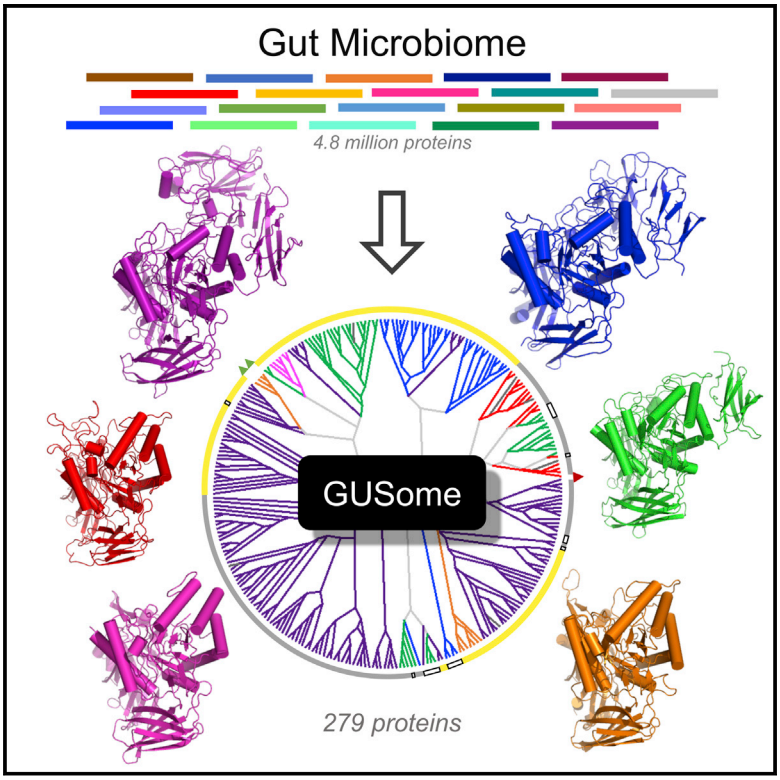
Article

Structure

An Atlas of β-Glucuronidases in the Human Intestinal Microbiome

Graphical Abstract



Highlights

- β-Glucuronidase (GUS) enzymes from the Human Microbiome Project examined
- 3,013 total and 279 unique microbial GUS proteins cluster in six structural groups
- GUSs are universally present and exhibit distinct inter-individual variabilities
- 112 novel GUSs are discovered, and functional differentiation is established

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In Brief

Microbial β-glucuronidase (GUS) enzymes cause drug-induced damage in the mammalian gut. We identify the 3,013 total and 279 unique microbial GUS proteins in the Human Microbiome Project database, and reveal their differential processing of distinct substrates. These data provide a roadmap for examining microbiome-encoded enzymes essential to human health.

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Structure
Article

An Atlas of β-Glucuronidases in the Human Intestinal Microbiome

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SUMMARY

Microbiome-encoded β-glucuronidase (GUS) enzymes play important roles in human health by metabolizing drugs in the gastrointestinal (GI) tract. The numbers, types, and diversity of these proteins in the human GI microbiome, however, remain undefined. We present an atlas of GUS enzymes comprehensive for the Human Microbiome Project GI database. We identify 3,013 total and 279 unique microbiome-encoded GUS proteins clustered into six unique structural categories. We assign their taxonomy, assess cellular localization, reveal the inter-individual variability within the 139 individuals sampled, and discover 112 novel microbial GUS enzymes. A representative in vitro panel of the most common GUS proteins by read abundances highlights structural and functional variabilities within the family, including their differential processing of smaller glucuronides and larger carbohydrates. These data provide a sequencing-to-molecular roadmap for examining microbiome-encoded enzymes essential to human health.

INTRODUCTION

The gastrointestinal (GI) microbiota is a highly complex ecosystem that lives in intimate symbiosis with its mammalian host (David et al., 2013; Kau et al., 2011; Maurice et al., 2013; Methé et al., 2012; Human Microbiome Project Consortium, 2012; Turnbaugh et al., 2008). Such complexity, however, makes probing the molecular basis of microbial-mammalian mutualism a daunting task. Mechanistic studies are typically reductionist, focusing on specific gene products and examining them in model systems. While next-generation sequencing continues to provide a wealth of paradigm-shifting data, its results are more difficult to reduce to a protein-level understanding of the molecular connections between the microbiota and the mammalian host.

Fortunately, microbial-mammalian symbioses often exist in interlocking functional systems that co-evolved for mutual benefit. These systems can in principle be reduced to a manageable number of molecular factors, allowing for the use of next-generation sequencing data to probe the interplay between the host and the GI microbiota. One such system is the inactivation of endobiotic and xenobiotic compounds by mammalian phase II drug metabolism and the subsequent reactivation of these chemicals by GI microbial enzymes (Figure 1A). The mammalian UDP-glucuronosyltransferase (UGT) family of drug metabolizing proteins link single glucuronic acid (GlcUA) sugars to a variety of compounds marked for excretion via the GI tract (Mackenzie et al., 1997; Meech et al., 2012). In the intestines, symbiotic bacteria express β-glucuronidase (GUS) enzymes capable of removing the inactivating GlcUA to use as a carbon source, and either further metabolize the parent compound or release it into the GI lumen (Mani et al., 2014). Previously glucuronidated compounds reactivated by GI microbial GUS enzymes can be reabsorbed to reach the plasma and be subject to cycles of enterohepatic recirculation (Figure 1A) (Redinbo, 2014).

A great deal is known about the diversity, activity, and expression of the mammalian UGTs (Meech et al., 2012). The UGT isoforms responsible for glucuronidation of a range of compounds including hormones, neurotransmitters, anti-cancer agents, non-steroidal anti-inflammatory drugs (NSAIDs), and environmental carcinogens have been characterized. By contrast, despite their importance in physiology and disease, relatively little is known about the GI microbial GUS enzymes that process endobiotic and xenobiotic glucuronide metabolites (Wallace et al., 2015). While microbial GUS activity has been demonstrated against many of the compounds known to be processed by mammalian UGTs, the microbes and associated enzymes responsible for this activity have not been elucidated. Microbial GUS activity is of particular interest because NSAIDs and the cancer drug irinotecan cause significant intestinal damage through the actions of GI microbial GUS enzymes (Boelsterli et al., 2012; Takasuna et al., 1996). These effects have been alleviated in mice using bacterial GUS-targeted inhibitors

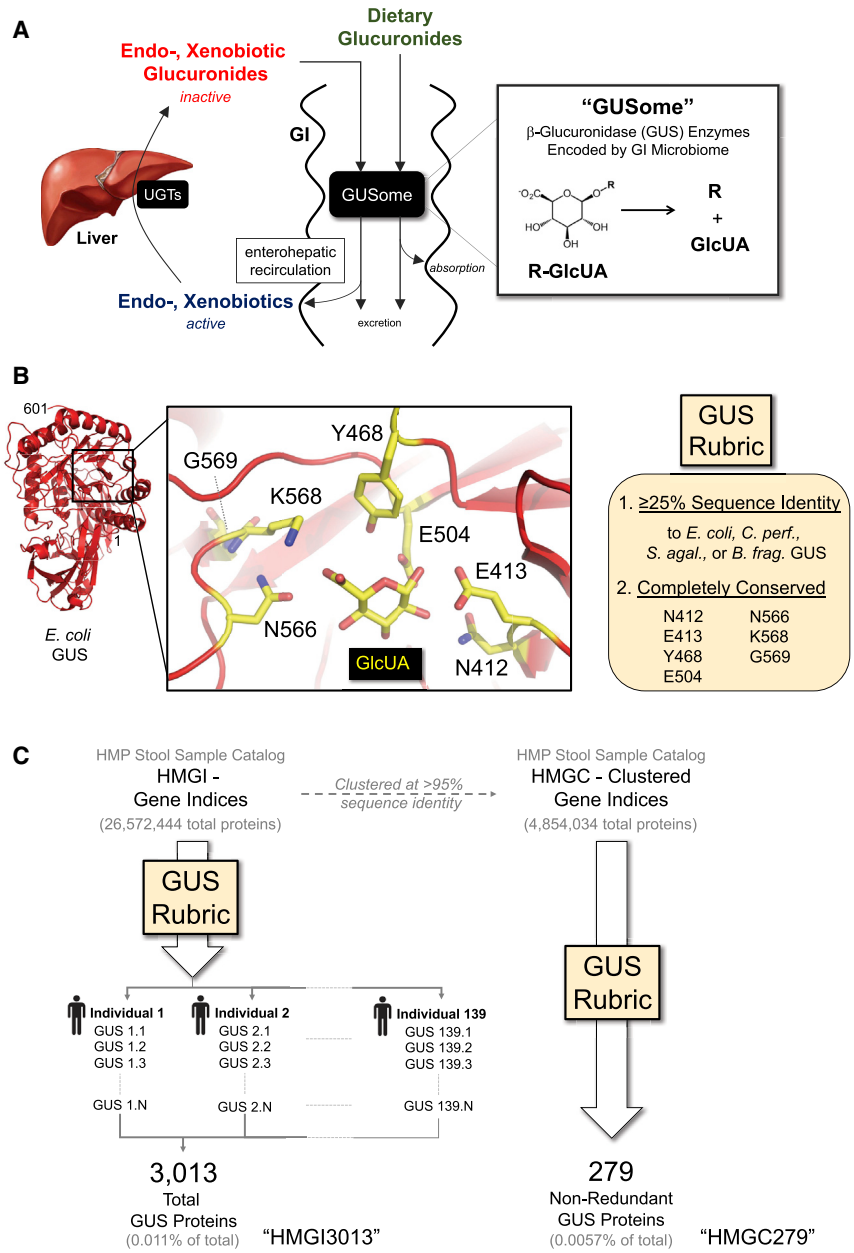


Figure 1. Human GI Microbial GUS Enzymes, the GUS Rubric, and the GUSome

(A) Compound inactivation by UDP-glucuronosyltransferases (UGT) that add glucuronic acid sugar (GlcUA). Endogenous (endobiotic), foreign (xenobiotic), and dietary glucuronides are substrates for the microbial β-glucuronidase (GUS) enzymes in the gastrointestinal (GI) tract, collectively termed the GUSome.

(B) *E. coli* GUS structure and active site residues employed to create the GUS Rubric.

(C) Two of the Human Microbiome Project (HMP) Stool Sample Catalog datasets were screened, with 3,013 GUS proteins identified from the HMGI (HMGI3013), a collection of all GUS proteins from 139 individuals, and 279 GUS proteins (HMGC279) from the non-redundant HMGC. See also Data S1, S2, and STAR Methods.

(LoGuidice et al., 2012; Mani et al., 2014; Roberts et al., 2013; Saitta et al., 2014; Wallace et al., 2010, 2015). In addition to their role in the UGT pathway of drug metabolism, GUS enzymes are predicted to metabolize dietary and host-generated polysaccharides containing terminal glucuronide moieties (Koropatkin et al., 2012; Salyers and O'Brien, 1980).

The specific types and diversity of GI microbial GUS enzymes responsible for processing these chemically distinct glucuronide substrates have not been determined, in part because we lack an understanding of the comprehensive diversity of bacterial GUS proteins present in the mammalian GI tract. The GUS enzyme was first described in 1934 (Masamune, 1934; Oshima, 1934), and GUS activity has since been noted in the major bacterial phyla of the mammalian GI: Bacteroidetes, Firmicutes, Proteobacteria, and Actinobacteria (Gloux et al., 2011; Kim et al., 2000; McBain and Macfarlane, 1998; Nakamura et al., 2002; Russell and Klaenhammer, 2001). Next-generation sequencing has also revealed novel β-glucuronidases, providing important advances in our understanding of unique GUS enzyme orthologs present in the mammalian GI (Gloux et al., 2011). Despite these data, however, a complete view of the GUS enzymes in the GI microbiome (a "GUSome") has yet to be assembled.

Here, we provide a comprehensive catalog of microbial GUS genes from gut metagenomic data assembled by the Human Microbiome Project (HMP). Guided by crystal structures of microbial GUS enzymes, we identify and describe the human GI microbial GUSome and characterize a representative set of GUS enzymes that sample its structural and functional diversity. These results facilitate a deeper understanding of glucuronide processing by GI microbes that affects host homeostasis and disease, and provide a roadmap for the creation of similar functional catalogs of other GI microbial protein families present in microbiome deep-sequencing data.

RESULTS

Identifying GUS Proteins from Human GI

The HMP Stool Sample Catalog, the set of genes sequenced from fecal samples collected by the HMP, contains more than 26.5 million translated protein sequences in its Gene Indices (HMGI) and 4.8 million in its Clustered Gene Indices (HMGC) databases (Methé et al., 2012). The clustered data consolidate proteins that share >95% sequence identity, and thus represent a library of unique proteins in the human GI microbiome. To find

genes that encode GUS enzymes in these vast databases, a two-step rubric was employed based on microbial GUS proteins of known structure and the conservation of residues essential to GUS function (Figure 1B). First, the HMGI and HMGC databases were screened to identify proteins with E values < 0.05 and sequence identity ≥25% in a pairwise alignment to at least one of the four microbial GUS enzymes of known crystal structure from *Escherichia coli*, *Clostridium perfringens*, *Streptococcus agalactiae*, and *Bacteroides fragilis* (Wallace et al., 2010, 2015). This step captured 1,646,315 proteins from the HMGI (6.2% of the total) and 267,594 proteins from the HMGC (5.5% of the total), and likely represent the wide range of microbial glycosyl α/β hydrolases. Although this type of information can be provided by the Pfam database (Finn et al., 2016), for example, which correctly identifies glycosyl hydrolase domains in GUS and β-galactosidase (GAL) candidate sequences, importantly, the Pfam database is not able to definitely distinguish between these two enzymes. GUS and GAL proteins share the same fold but maintain functionally critical differences in their active sites. For example, we previously identified the "N-K motif" unique to the active site of GUS enzymes and distinct from GAL proteins, and used that information to correct a microbial GUS initially classified as a GAL (3CMG from *B. fragilis*) (Wallace et al., 2015).

Second, the 1,646,315 and 267,594 sequences identified from the HMGI and HMGC databases, respectively, in the first step were screened for conserved active site residues essential to hydrolyzing glucuronide-containing substrates (Figure 1B) (Wallace et al., 2010, 2015). The conserved residues included the two catalytic glutamic acids and a proximal asparagine as well as invariant Gly, Asn, and Lys side chains (N566, K568, G569; the "N-K motif" noted above) that distinguish the carboxylic acid moiety unique to GlcUA relative to the primary alcohol in the nearly isostructural galactose (Figure 1B) (Matsumura and Ellington, 2001; Wallace et al., 2015). These GUS-specific active site residues were identified using a combination of structural and functional data (Matsumura and Ellington, 2001; Roberts et al., 2013; Wallace et al., 2010). Furthermore, the seven residues selected to define the GUS rubric (Figure 1B) were completely conserved in multiple sequence alignments in confirmed GUS enzymes and in GUS proteins of known structure (Callanan et al., 2007; Gloux et al., 2011; Hassan et al., 2013; Wallace et al., 2010, 2015).

The second, active site structure-guided step identified 3,013 GUS proteins (0.011% of the total) from the full 26.5 million member HMGI database (we term this group HMGI3013; Figure 1C; Data S1), and 279 unique non-redundant GUS proteins (0.0057% of the total) from the 4.8 million HMGC database (termed HMGC279; Figure 1C; Data S2). Structure-guided screening reduced the number of glycosyl hydrolase fold-containing proteins selected from the first step by three orders of magnitude. These data define the first collective human GI microbial GUSome. We expect that only GUS-type enzymes were identified because no validated GAL was present in the HMGI3013 or HMCG279 datasets created. Furthermore, because the HMGI3013 dataset was assembled from 139 individuals, the number and types of GUS enzymes present within each individual can be examined (e.g., GUS1.1, the first GUS from individual 1, etc.; Figure 1C; STAR Methods). These data significantly extend our understanding of the catalytic potential of the microbiota and demonstrate the utility of employing structure-guided rubrics to annotate microbiome sequencing data.

Distinct Enzyme Architectures in the GUSome

While hundreds to thousands of proteins are more tractable than millions, we sought to further categorize the GUSome. In multiple sequence alignments of the HMGC279 proteins, the active site residues and core predicted α/β hydrolase folds were highly conserved (Figure 2A; Data S3). By contrast, the active site-adjacent loop 1 (L1) and loop 2 (L2) exhibited marked variability in length and amino acid composition (Figures 2A and 2B; Data S3). These regions were therefore employed to generate more manageable categories of GUSome proteins (STAR Methods; Data S3). L1 was initially termed the "Bacterial Loop" in *E. coli* GUS (Figure 2A) (Wallace et al., 2010); however, as outlined below, L1 exhibits two sizes: a full L1 of >15 residues and a smaller mini-loop 1 (mL1) of 10–15 residues. The second region, L2, is adjacent in structure to L1, located three residues in the amino acid sequence away from a catalytic glutamic acid, and is variable in length across the GUSome (Figures 2A and 2B; Data S3). Like L1, L2 occurs in a full size of ≥12 residues, as well as a mini-loop 2 (mL2) of 9–11 residues.

Each protein in the HMGI3013 and HMGC279 GUSome datasets can be classified into one of seven categories: L1, mL1, L2, mL2, mini-loop 1,2 (mL1,2), no loop (NL), and no coverage (Figures 2B and 2C; Tables 1, S1, and S2). "No coverage" is not a formal enzyme category; instead, these predicted proteins lack sequencing information for one or both loop regions. They are a small fraction (2.2%–4.8%) of the GUS protein sets in-hand (Tables 1 and S2; Data S2; Figure S1) and are indicative of sequencing "dark matter" present in whole-genome metagenome data.

L1 GUS enzymes contain a >15-residue loop at the 356–380 region of *E. coli* GUS (Figures 2B and 2C; Data S3) (Wallace et al., 2010). Additional L1 GUS structures from the Firmicutes species *C. perfringens* and *S. agalactiae* contain 22- and 21-residue loops in this location (Data S3) (Wallace et al., 2015). The L1 regions of Firmicutes *Eubacterium eligens* and *Faecalibacterium prausnitzii* GUS enzymes identified in this study (see below) reveal 24- and 27-residue loops (Figures 2B and 2C; Data S3). L1 enzymes represent ~5.5% of the GUS protein sequences identified (Tables 1 and S2). mL1 enzymes contain a loop of 10–15 residues in the same location as L1. *B. fragilis* GUS of the Bacteroidetes phylum, with a 12-residue loop, is an mL1 GUS of known structure (Figures 2B and 2C; Data S3; PDB: 3CMG). mL1 enzymes are relatively abundant, represent 15%–21% of the total GUS enzymes identified (Tables 1, S1, and S2), and can contain additional C-terminal carbohydrate-binding domains, as outlined below.

L2 GUS proteins contain a ≥12-residue insertion region 416–419 of *E. coli* GUS, and are represented by a *Bacteroides uniformis* GUS (Figures 2B and 2C; see below). A model of *B. uniformis* GUS suggests a 19-residue loop in this position (amino acids 428–446; Figure 2C; Data S3). L2 GUS enzymes lack L1 motifs (Figures 2B and 2C; Data S3), represent 12%–14% of the GUS proteins identified (Tables 1, S1, and S2), and contain additional C-terminal domains relative to L1 enzymes. mL2 enzymes exhibit a loop of 9–12 residues in length in the same location,

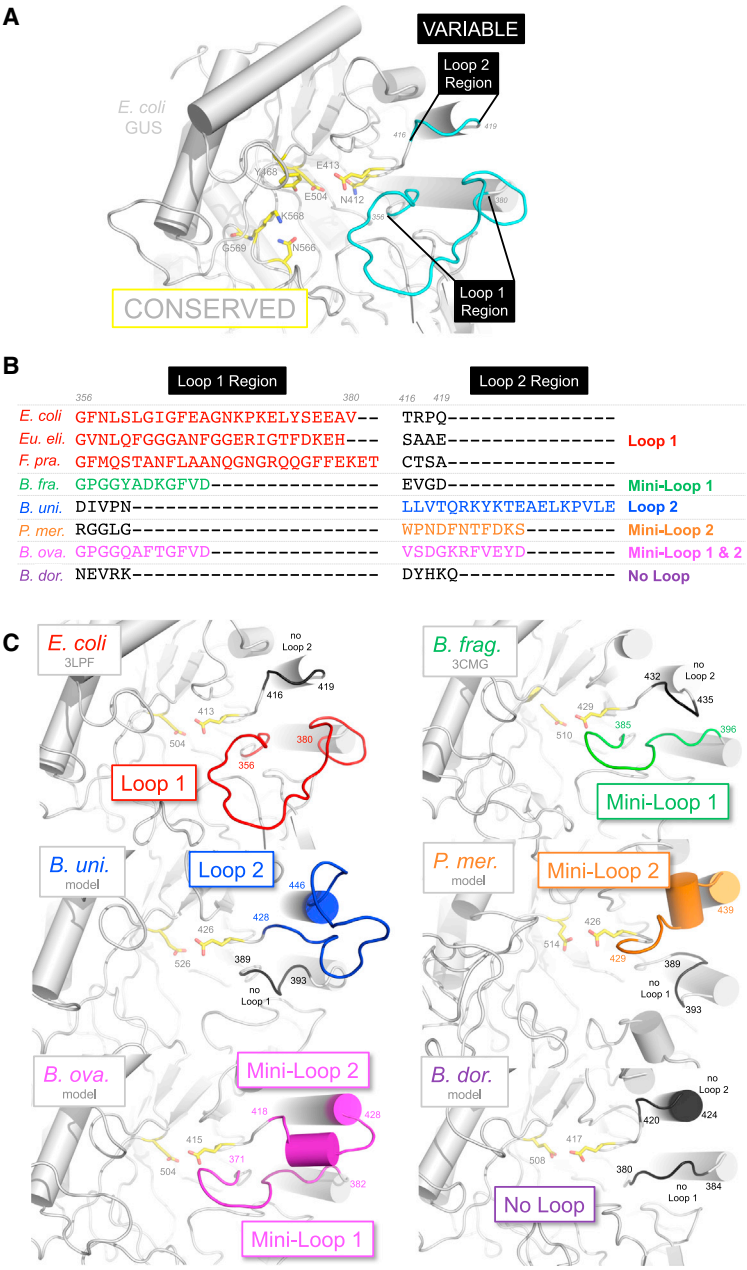


Figure 2. Human GI Microbial GUS Loop Categories

(A) *E. coli* GUS structure (gray) with conserved active site (yellow) and the variable Loop 1 and Loop 2 regions (cyan).

(B) Loop 1 and 2 sequences of representative GUS enzymes (*Eu., Eubacterium; F., Faecalibacterium; B., Bacteroides; P., Parabacteroides*) with residue numbers for *E. coli* GUS.

(C) Active sites of GUS enzymes from each loop category. *E. coli* and *B. fragilis* GUS are of known structure, while the others are models. See also Data S3 and S4.

contain no L1, and are represented by a *Parabacteroides merdae* GUS (Figures 2B and 2C; Data S3). mL2 proteins comprise 4%–7% of the total GUS enzymes in our analysis (Tables 1, S1, and S2).

A small number of GUS proteins, representing only 1%–2% of the total datasets, contain both mL1 and mL2 motifs (Figures 2B and 2C; Tables 1, S1, and S2). mL1,2 enzymes exhibit 10–15 residues in the L1 and 9–12 residues in L2 (Figures 2B and 2C; Data S3). A model of a representative GUS from *Bacteroides ovatus* reveals the two loops in close proximity and suggests that full loops in either location would clash, perhaps explaining why enzymes containing full L1 and L2 inserts were not observed (Figure 2C). Finally, NL GUS enzymes lack a loop in either position (Figures 2B and 2C; Data S3). NL enzymes are by far the most abundant GUS identified, comprising 49%–57% of each of our datasets (Tables 1, S1, and S2). The H11G11 GUS previously discovered from human fecal metagenomics data was identified in this HMP analysis and categorized as an NL enzyme (Gloux et al., 2011) (Data S3; Table S2). Taken together, the human GI microbial GUSome is composed of enzymes that decorate a conserved fold with distinct active site-adjacent loops, likely enabling them to act on chemically variable substrates present in the intestinal tract.

HMGC279 Taxonomy and Localization

Taxonomy was assigned and a cladogram based on sequence similarity was created for the HMGC279 non-redundant set of GUS enzymes (Figure 3A). Phylum-level taxonomy was fixed for 93.5% (261) of the 279 GUS enzymes identified, while 18 enzymes remain "unclear" due to moderate sequence identity to enzymes encoded by bacteria from multiple phyla (Figures 3A and 3B; STAR Methods; Table S2). Genus and species assignments were also possible for 177 of the 261 sequences of known phyla based on exhibiting ≥95% identity to a protein in the NCBI PDB (Table S2; Data S2). However, 112 GUS enzymes, 18 without clear phyla and 94 without genus and species, were not previously documented in NCBI. Thus, 40% of the 279 non-redundant GUSome set represent novel microbial GUS proteins (Table S2), an indication of the potential of the GI microbiome for enzyme discovery.

For the 261 sequences with unambiguous taxonomy, the following phyla were observed (Figures 3A and 3B; Table S2): Bacteroidetes (52%), Firmicutes (43%), Verrucomicrobia (1.5%), and Proteobacteria (0.5%). The Bacteroidetes phylum exhibits the broadest diversity and the largest number (137) of non-redundant GUS enzymes (Figures 3A and 3B; Table S2; Data S2). The Bacteroidetes contain proteins in five of the six loop categories, and is the only phylum that expresses the rare mL1,2 enzymes. However, no L1 enzymes were detected in the Bacteroidetes phylum (Figure 3A). The Firmicutes phylum has the next most abundant number of non-redundant GUS enzymes (112); they do not exhibit mL2 or mL1,2 enzymes, but do contain the majority of L1 and NL enzymes, a large fraction of mL1 enzymes, and one L2 enzyme (Figures 3A and 3B). The Verrucomicrobia phylum represents a small number of total GUS proteins (4), all of which are mL2, while the Proteobacteria phylum contains

Table 1. Composition of the HMGI3013 and HMGC279 GUS Enzyme Datasets by Loop Category

	Loop 1 (L1)	Mini-Loop 1 (mL1)	Loop 2 (L2)	Mini-Loop 2 (mL2)	Mini-Loop 1, 2 (mL1,2)	No Loop (NL)	No Coverage
HMGI3013 Composition, % (number)	5.6 (165)	20.7 (615)	11.9 (351)	6.7 (208)	1.3 (36)	49 (1,494)	4.8 (144)
HMGC279 Composition, % (number)	5.7 (16)	14.7 (41)	14 (39)	3.9 (11)	2.2 (6)	57.3 (160)	2.2 (6)
Length range number residues	595–611	578–685	830–1,085	685–807	661–665	555–1,059	NA
Examples	*E. coli, Eu. eligens, F. prausnitzii*	*B. fragilis*	*B. uniformis*	*P. merdae*	*B. ovatus*	*B. dorei*	NA

The "no coverage" category are proteins with no sequence coverage for either the Loop 1 or Loop 2. NA, not applicable.

a single GUS in the 279 set, the L1 enzyme from *E. coli* (Figures 3A and 3B).

We next sought to understand the cellular localization of members of the HMGC279 non-redundant GUS set by searching for an N-terminal signal sequence that would mark the enzyme for trafficking to the periplasmic space (STAR Methods) (Navarre and Schneewind, 1999). Predicted signal sequences were based on signal peptidase I sites and were not followed by a cysteine, suggesting that enzymes with signal sequences are retained in the periplasmic space (Auclair et al., 2012; Zückert, 2014). While 20% of the overall sequences could not be classified due to missing or miscalled starting methionines, clear trends emerged (Figure 3C; Table S2). No L1 enzyme contained a signal sequence, suggesting that L1 proteins are maintained intracellularly. By contrast, all L2, mL2, and mL1,2 enzymes contained signal sequences, indicating that these proteins can be transported across the inner microbial membrane. Roughly half of the mL1 and NL GUS proteins contained signal sequences, and these results were segregated by phylum: all Firmicutes mL1 and NL proteins lacked signal sequences, while all such Bacteroidetes enzymes contained them. These results suggest functional specialization, with periplasmic proteins possibly acting on larger polysaccharide substrates, while intracellular enzymes process smaller glucuronides capable of reaching this location within the cell.

Predicted signal sequences, if present, were trimmed from the proteins in the HMGC279 to define expected enzyme lengths; 61 proteins of the 279 have unconfirmed lengths because a clear translation start or end site could not be identified (STAR Methods; Figure 3D; Table S2). The L1, mL1, L2, mL2, and mL1,2 enzymes tended to cluster together by length classifications, while the NL enzymes sample the full range observed, from the smallest GUS at 555 amino acids to nearly the largest at 1,059 residues (Figure 3D). The increase in GUS enzyme length likely corresponds to the addition of protein domains to the C terminus of the conserved core α/β hydrolase fold, as demonstrated in more detail below. This conclusion is supported by the analysis of the HMGC279 sequences using the Carbohydrate Active enZymes (CAZy) database (Lombard et al., 2014), which describes structurally related proteins, including glycosyl hydrolases, that act on carbohydrates. CAZy classifications for the HMGC279 sequences with confirmed lengths reveal that all enzymes have conserved glycosyl hydrolase 2 (GH2) domains located N-terminally, while those that extend beyond that domain are commonly assigned a carbohydrate-binding module (CBM) domain located C-terminally (Figure 3D; Table S2). Together, these results reveal the diversity, localization, and functional specialization of the GUS enzymes in the human intestinal microbiome.

Inter-Individual Variability in HMGI3013

The >26 million-protein HMGI database is organized by samples obtained from 139 donors, allowing us to examine the GUS composition, distribution, and abundance within this set of healthy fecal donors (Figure 1C; Data S1; Tables 1 and S1). Our analysis only considers the presence or absence of GUS genes within individuals; however, not their expression or activity, which would require transcriptomics and/or proteomics investigations. First, the total number of GUS protein sequences in each individual's gut microbiota was determined. All samples contained a gene predicted to encode a GUS, with a minimum of 4 GUS proteins, in one individual, and a maximum of 38 GUS proteins, in two individuals; the mean number of GUS enzymes per individual sampled was 21.7 (Figure 4A). Thus, while GUS proteins were relatively rare within the overall microbiome (e.g., 0.006%–0.01% of total genes; see Figure 1C), they were universally associated with a healthy microbiota in this database.

Second, each GUS in the HMGI3013 was classified into a loop category, and the number of enzymes in each category were compared across individuals (Tables 2 and S1). Only NL GUS enzymes were observed to be present in all 139 samples, although two individuals had only one NL GUS, while one individual had 23 (Table 2; Figure S1). The mean number of NL enzymes per sample was 10.7, supporting the conclusion that NL enzymes represent the majority of GUS proteins in the human GI microbiome (Table 2; Figure S1). mL1 GUS enzymes were present in 99% of individuals, with a maximum of 11 and a mean of 4.4 per sample. L2 and mL2 GUS proteins were present in 92% and 81% of individuals, respectively. We also considered the average composition of each GUS type per individual including only samples which contained at least one of these GUS classifications; thus, for L2 and mL2, the average values increase from 1.6 and 1.1 to 2.8 and 1.9 per individual, respectively (Table 2). While NL, mL1, L2, and mL2 enzymes are relatively abundant, L1 and mL1,2 enzymes are less common. L1 enzymes are only present in two-thirds of individuals. mL1,2 are rare, present in less than 25% of individuals and exhibit a mean of only 1 when they are present (Table 2; Figure S1). Taken together, these

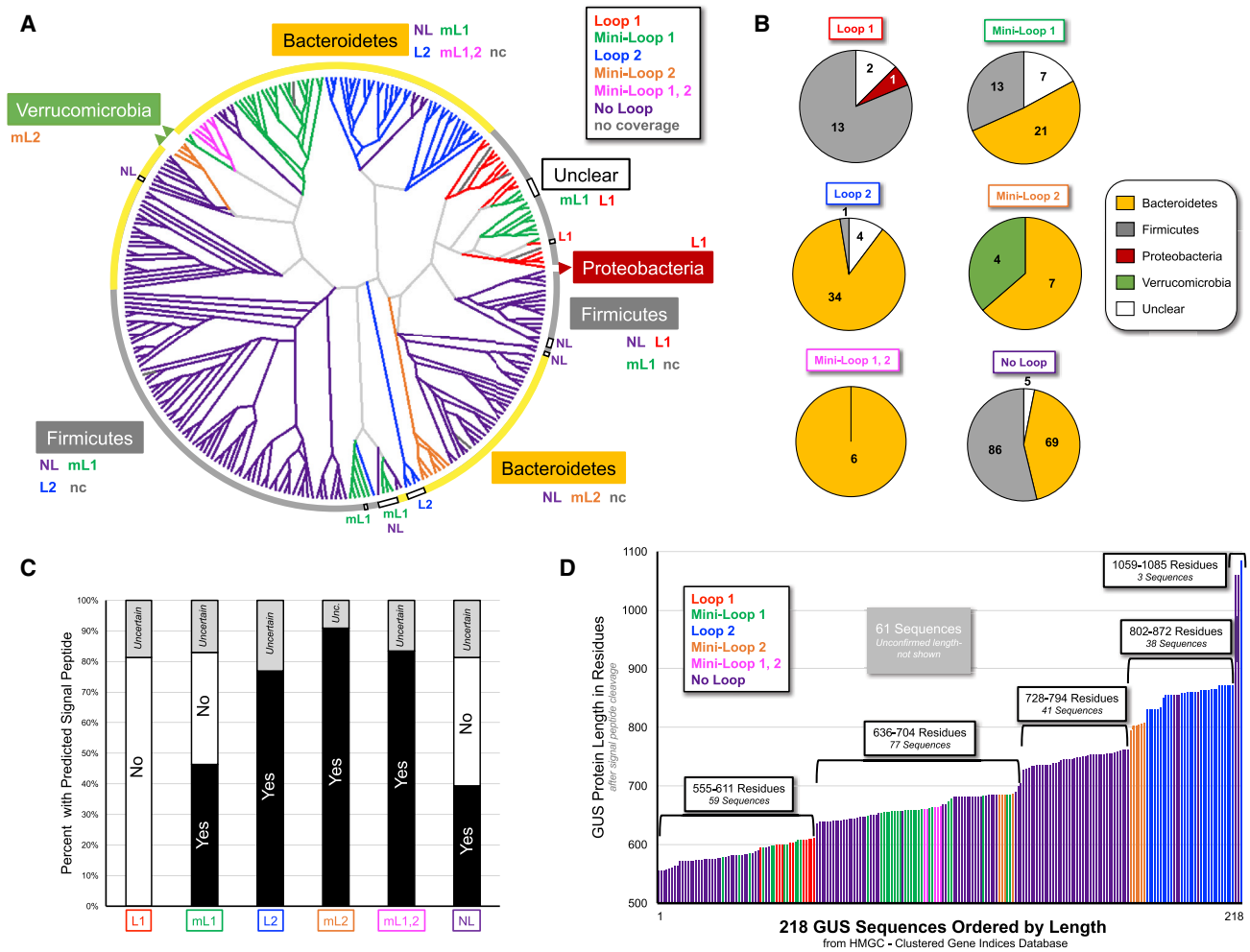


Figure 3. Human GI Microbial HMGC279 Taxonomy, Localization, and Composition

(A) Cladogram for the 279 GUSome including a few without clear taxonomy (unclear). GUS loop types in each phylum are indicated.

(B) Phylum composition with numbers of GUS enzymes per phylum indicated.

(C) Predicted signal sequence prevalence for each GUS type. GUS without clear signal sequences are thus labeled "uncertain."

(D) GUS protein length colored by loop category. Sixty-one had unclear sequences and were not assigned a length.

data reveal the compositional variability of this microbial enzyme in the human GI, and will facilitate future studies designed to test the functional variability across individuals.

The HMGI database is comprised of whole-genome metagenome deep-sequencing reads. To gain an appreciation of the relative abundances of each GUS type, read abundances were calculated per sample and normalized to allow comparison across individuals (STAR Methods; Table 2; Figure S1). The overall results reflect those in Tables 1 and 2, with NL enzymes being most abundant, followed by mL1, L2, mL2, L1, and mL1,2 (Figures 4B and S2). However, marked variabilities were noted across individuals (Figure S1; Table S1). For example, some individuals contained much higher than average abundances of mL1 and L2 enzymes, while others showed increases in L1 proteins (Figure 4B). Indeed, one individual's sample contained 25% mL1,2 enzymes, by far the least common GUS by other measures (Figure 4B; see also Figures 3A and 3B; Tables 1 and S1). Such variability may be stochastic, or may reflect important functional differences between the metabolic capabilities of the microbiota in these individuals. No information on diet, health, or therapeutic use, for example, is available for the 139 individuals in the HMP Stool Sample Catalog; thus, resolving the basis for these distinctions will require future studies with matched metadata.

GUSome Structure and Function

To generate a representative set of microbial GUS proteins for in vitro study, the most common GUS protein of each category was identified from the HMGI3013 dataset. This was achieved by tabulating the most abundant GUS of each category within each individual by normalized read counts (Figure 4C, step 1) and then tabulating across individuals (Figure 4C, steps 2 and 3; STAR Methods). Bacteroidetes and Firmicutes dominate these abundance results, with the highly abundant enzymes in each category coming from these two phyla (Figure 4D). The exception is the mL2 proteins from the Verrucomicrobia phylum

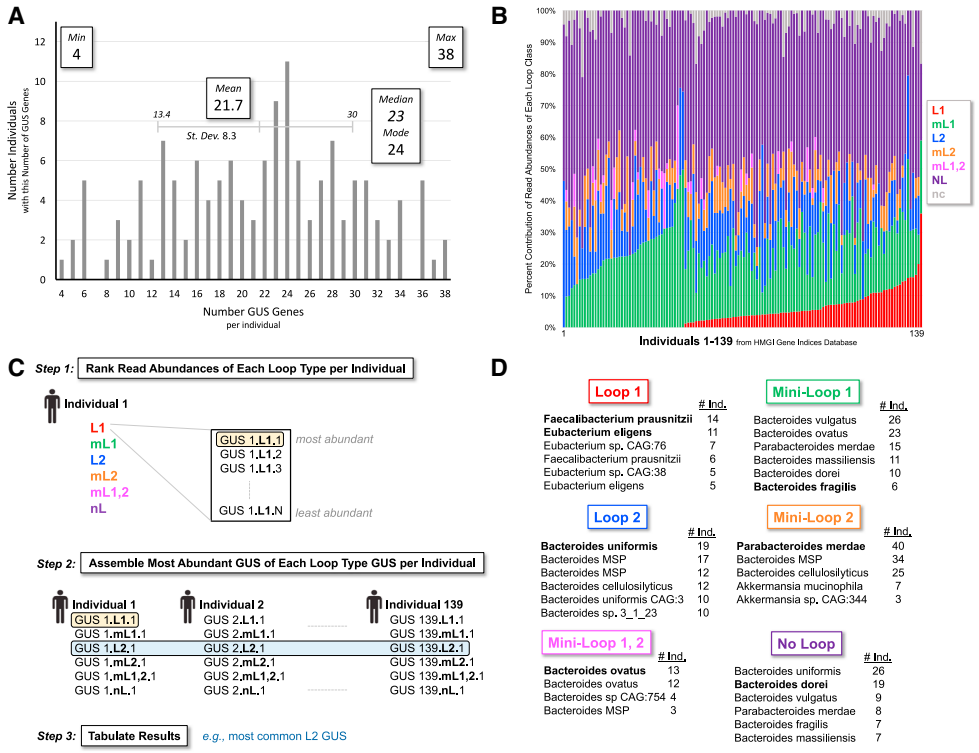


Figure 4. Human GI Microbial HMGI3013 Composition and Abundances

(A) The 139 individuals in the HMGI contained 4 to 38 GUS proteins per individual.

(B) GUS composition by read abundance in each individual and rank-ordered by L1.

(C) Stepwise use of read abundances to identify the most common GUS of each loop category. The results of step 3 are presented in (D).

(D) Abundant GUS enzymes in each category, with those characterized further in bold. See also Figures S1 and S2.

(genus *Akkermansia*; Figure 4D). These results provide a first view of the GUS enzymes that govern the processing of glucuronides in the GI tract, and provide a rationale for choosing novel GUS enzymes for in vitro characterization.

Indeed, a representative panel of novel GUS enzymes was created for in vitro analysis. The enzymes highlighted in bold in Figure 4D were purified to homogeneity (STAR Methods). Six GI microbiome-encoded GUS enzymes were chosen from the HMP analysis as well as two GUS enzymes already in-hand, the L1 *E. coli* GUS (EcGUS) (Wallace et al., 2010) and the mL1 GUS from *B. fragilis* (BfGUS). BfGUS has recently been shown to be a GUS (Wallace et al., 2015), is of known structure (PDB: 3CMG), and shares 66% sequence identity with the top mL1 proteins identified. Of the novel enzymes chosen from the HMP abundancy analysis, two were L1 enzymes, from *F. prausnitzii* (FpGUS) and *Eu. eligens* (EeGUS; Figure 4D). The other four represent one each from the remaining categories: L2 from *B. uniformis* (BuGUS; Figure 4D), mL2 from *P. merdae* (PmGUS), mL1,2 from *B. ovatus* (BoGUS), and NL from *B. dorei* (BdGUS). The eight proteins in this panel share between 21% and 45% sequence identity (Figures 5A and S3), and range between 603 and 886 residues in length (Figure 5A).

The representative eight-protein GUS panel was first examined with the standard GUS assay substrate *p*-nitrophenol glucuronide (PNPG), a small glucuronide (Figure 5A; STAR Methods). Activity optima were determined for each enzyme within the pH range of 4–7.4 (Figures 5A and S4), and noted differences were observed. Within L1 proteins, EcGUS is active across a broad pH range, including pH 7.4, while EeGUS and FpGUS perform best at 6.5 and 6, respectively (Figures 5A and S4). Such differences may reflect the relative locations of microbial species within the GI, as the pH gradually increases from ~5 in the proximal duodenum to ~7.4 in the sigmoid colon (Abuhelwa et al., 2016). PNPG kinetic studies with each enzyme at its optimum pH (STAR Methods) revealed apparent catalytic turnover rates for the L1, mL1, and L2 enzymes to be much higher than the mL2, mL1,2, and NL GUS enzymes tested (Figure 5A). Indeed, within the L1 enzymes, *E. coli* and *Eu. eligens* GUS are remarkably efficient at PNPG processing, exhibiting rates 10-fold higher than the L1 protein from *F. prausnitzii,* and similarly higher than the mL1 and L2 enzymes from *B. fragilis* and *B. uniformis*, respectively (Figure 5A). This may be due to the L1 region, which could be capable of closing over small substrates within the GUS active site (see Figures 2A and 2C) (Wallace et al., 2010, 2015).

L2 GUS proteins also maintain a reasonably long, >15-residue amino acid region located adjacent to L1, and thus could also close over the catalytic site in a manner similar to L1 GUS enzymes. To understand L2 GUS enzymes, we determined the 1.9 Å resolution crystal structure of *B. uniformis* GUS (Figure 5B; Table S3). Contrary to expectations, the 19 amino acids of BuGUS's L2 form a helix-turn-helix motif rather than a loop; thus, the active site of the L2 BuGUS in this structure is relatively

Table 2. Inter-Individual Variability in the Composition of GUS Types in HMGI3013 Dataset

	Loop 1 (L1)	Mini-Loop 1 (mL1)	Loop 2 (L2)	Mini-Loop 2 (mL2)	Mini-Loop 1, 2 (mL1,2)	No Loop (NL)	No Coverage
Individuals with at least one of this GUS type, % (number out of 139 total)	66 (92)	99 (137)	92% (128)	81% (112)	24% (34)	100% (139)	59% (82)
Range of number GUS of this type per individual	0–5	0–11	0–7	0–4	0–2	1–23	0–5
Average number per individual, average (SD)	1.2 (1.2)	4.4 (2.2)	2.5 (1.6)	1.5 (1.1)	0.25 (0.5)	10.7 (4.6)	1.0 (1.2)
Average number per individual with at least one of this GUS type, average (SD)	1.8 (1.0)	4.5 (2.2)	2.8 (1.5)	1.9 (0.8)	1.1 (0.2)	10.7 (4.6)	1.8 (1)

open compared with the active sites of L1 enzymes (Figure 5B). The BuGUS L2 structure also reveals additional domains added to the C terminus of this 886-residue enzyme compared with the smaller, ~600-residue L1 GUS proteins. BuGUS contains two carbohydrate binding modules (CBMs) (Boraston et al., 2004) that are missing from L1 GUS enzymes. The BfGUS mL1 enzyme, by contrast, contains one C-terminal CBM (Figure 5C). Interestingly, both the BfGUS CBM and the first, more N-terminal CBM of BuGUS are not detected as CBMs in CAZy database searches (Lombard et al., 2014), suggesting that these represent previously uncharacterized CBMs. Thus, the increased GUS protein lengths presented in Figure 3D can, in these cases, be explained by additional CBMs to facilitate the binding of larger substrates, possibly carbohydrates.

Finally, we hypothesized that microbial GUS enzymes with open active sites would process polysaccharides with a GlcUA at the non-reducing end. We tested this hypothesis by examining our panel of eight GUS enzymes with a representative natural heparan nonasaccharide: GlcUA-(GlcNAc-GlcUA)$_4$-PNP (GlcUA, glucuronic acid; GlcNAc, N-acetylglucosamine; PNP added for detection) (STAR Methods). We found that L1 GUS enzymes were not able to process this substrate, yielding no product over 24 hr (Figure 5A). By contrast, the mL1, L2, mL2, and NL enzymes were able to utilize this heparan substrate, with the mL2 GUS from *P. merdae* converting 100% of it to product (Figure 5A). Analysis of products by mass spectrometry revealed that only the terminal glucuronic acid was removed by each GUS (Figures 5A and S5). The mL1,2 GUS from *B. ovatus* showed no activity with this heparan substrate (Figure 5A), indicating that we have yet to identify the substrate of this unique and relatively rare type of GUS. Taken together, these data on representative GUS enzymes provide a critical first view of the structural and functional diversity present in this microbiome family of proteins.

DISCUSSION

An atlas of human GI microbiome-encoded enzymes comprehensive for the HMP Stool Sample Catalog is presented. The results demonstrate conservation of structural features in bacterial GUS proteins and their diversity in function, structure, and cellular localization. The GUSome enzymes sample four major microbial phyla (Bacteroidetes, Firmicutes, Verrucomicrobia, and Proteobacteria), and can be clustered into six distinct categories. GUS enzymes from these categories are also shown to exhibit differing functional capacities; enzymes with longer loops process a small glucuronide substrate, while others with open active sites process a larger heparan substrate. GUS proteins active on small substrates appear to be intracellular, while enzyme orthologs that act on large polysaccharides maintain signal sequences and are expected to be localized in the periplasmic space (Koropatkin et al., 2012).

Connecting phyla with GUS functional patterns, we hypothesize that NL enzymes are housekeeping proteins maintained by a large number of Firmicutes and Bacteroidetes taxa for the processing of commonly available substrates, including host- or microbial-produced polysaccharides, such as heparan, heparin, chondroitin, and mucins. Such housekeeping NL enzymes can be either intracellular or extracellular. While we show they can act on one type of polysaccharide, there is an array of important polysaccharides present in the GI lumen (Koropatkin et al., 2012). We hypothesize that distinct NL proteins, and also mL1, L2, mL2, and mL1,2 GUS enzymes, will show preferences for different types of polysaccharide substrates. Future studies will be required to assign substrate preferences to each category of GUS proteins, and perhaps even to individual GUS enzymes within each category, to understand fully the roles these proteins play in mammalian-microbial mutualism.

Intracellular GUS enzymes appear poised to respond to the presence of smaller glucuronide conjugates formed either locally by the GI epithelium, which express UGTs, or by the liver, which passes glucuronide conjugates to the GI via the bile duct (Mackenzie et al., 1997; Meech et al., 2012). Such substrates are expected to exhibit diverse aglycone chemistry, sampling the sterol scaffold compounds and bilirubin for endobiotics and, for xenobiotics, a broad range of drugs and toxins (Meech et al., 2012). Such intracellular systems would be expected to require importers to bring small glucuronide conjugates into microbial cells for processing. Indeed, the Enterobacteriaceae are known to encode a GUS operon that expresses both the GUS enzyme (*gusA*), as well as inner- and outer-membrane glucuronide mobilization proteins (*gusB* and *gusC*, respectively) (Liang et al., 2005; Wilson et al., 1992). The GI microbiota harbor an array of distinct metabolic functions that can be increased both by gene expression and by clonal expansion when particular substrates are present, and then decreased by relative growth disadvantages when key compounds are scarce.

The current report details 112 new GUS proteins that lack identifiable phylum or genus, demonstrating that precise

A

GUS	Length (residues)	pH Optimum	PNPG Cleavage Rate (nmol PNP / min / nmol GUS)	Heparan Cleavage (% in 24 hr)
E. coli	603	7.4	7400 ± 200	0
Eu. eli.	611	6.5	9400 ± 500	0
F. pra.	608	6	650 ± 4	0
B. fra.	693	5	560 ± 20	37 ± 10
B. uni.	886	5.5	540 ± 20	71 ± 3
P. mer.	826	5.5	1.5 ± 0.07	100 ± 0
B. ova.	686	5	2.0 ± 0.07	0
B. dor.	875	6	98 ± 2	37 ± 9

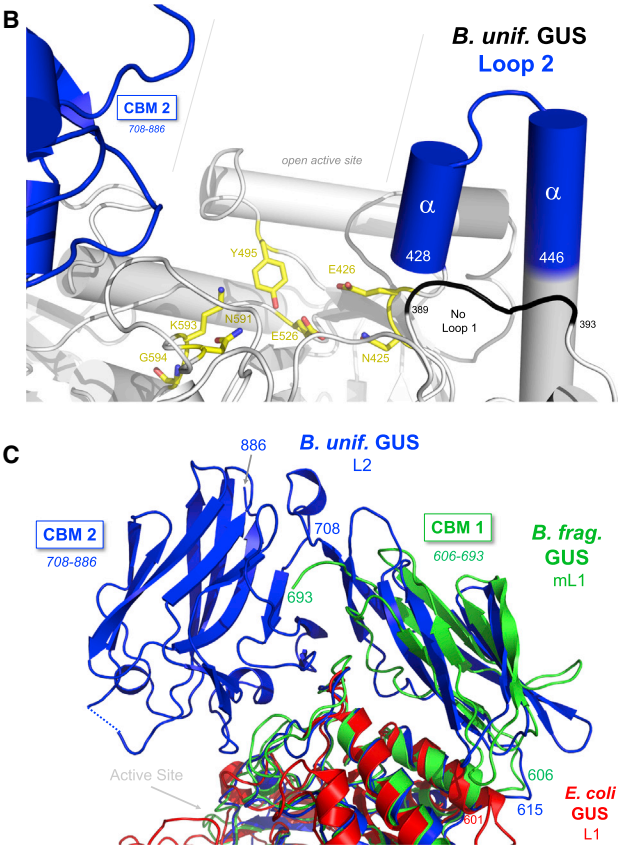


Figure 5. Human Microbial GUS Functional and Structural Specialization

(A) Differential processing of a small *p*-nitrophenol glucuronide (PNPG) substrate and a larger heparosan nonasaccharide substrate composed of GlcUA-(GlcNAc-GlcUA)$_4$-PNP (GlcUA, glucuronic acid; GlcNAc, N-acetyl-glucosamine; labeled "heparan"). Rates of PNPG processing are nmol PNP produced per min per nmol GUS, while percent conversion of the heparan substrate over 24 hr is provided. SE is reported for each.

(B) Active site of the crystal structure of the Loop 2 GUS from *B. uniformis* (PDB: 5UJ6). Conserved catalytic residues (yellow), helix-turn-helix motif (blue), and the carbohydrate-binding region (CBM 2; blue) shown.

(C) The CBM 1 of Mini-Loop 1 GUS from *B. fragilis* (3CMG; green) aligned with the longer region of BuGUS with CBM 2 (blue). *E. coli* Loop 1 GUS (red; 3LPF) lacks a CBM at its C terminus.

structure-guided searching of metagenome data has the capability of expanding our knowledge of microbial functions by 40%, in this case, and perhaps more for other catalytic functionalities. Moreover, this strategy can be extended to enzymes other than GUS given defined structural motifs to identify them, enabling the mining of whole-genome metagenome sequences from the GI as well as other body sites. However, a limitation to the study presented here is the requirement that the GUS enzymes selected maintain all seven of the conserved active site residues shown in Figure 1B. It is likely, given the rapid evolutionary capability of bacteria, that the GI microbiota have sampled active sites with one or more of these residues altered to orthologous or distinct amino acid side chains, and yet still maintain GUS activity. Future studies will be required to examine this possibility. A second limitation to the current study is that, despite the GUS rubric employed, there may be enzymes we have labeled here as a GUS that in fact lack GUS activity. For this reason, we have chosen to describe this is "an atlas" of GUS enzymes, rather than the definitive atlas, and we look forward to further studies designed to probe this first atlas for functional diversity and potential surprises.

The 3,013 GUS proteins identified from the fecal sampling of 139 individuals provide the first full-scope view of GI microbial enzyme diversity and compositional heterogeneity in humans. All individuals contained GUS-encoding genes, but the numbers varied by 10-fold across individuals, from a minimum of 4 to a maximum of 38 GUS proteins. The functional diversity was also observed to be relatively vast, with some individuals encoding a large number of relatively rare GUS enzymes, such as L1 or mL1,2, compared with most samples that exhibited dominance of NL enzymes. More detailed knowledge about whether unique GUS compositions are stochastic or representative of other factors such as diet or disease will remain for future studies. Indeed, the data from these 139 individuals, while well-annotated, are only a small first look at the diversity of a family of gut microbiome-encoded enzymes, and we look forward to future studies of larger metagenomic assemblies collected from more individuals and integrated with matched metadata.

The information outlined here could form a basis for personalized medicine and predictive efforts to identify patients at risk for chemotherapy-induced diarrhea, for example. Such knowledge may facilitate regimen-tailoring to enhance anti-tumor efficacy while avoiding dose-limiting GI toxicities, or could suggest dietary changes that could improve microbiota composition before treatment. Taken together, the data and methodologies presented provide a roadmap for the annotation of diverse microbiome-encoded catalytic and non-catalytic functions, and open the door to new eras of discovery in understanding mammalian-microbial mutualism.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

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● DATA AND SOFTWARE AVAILABILITY

SUPPLEMENTAL INFORMATION

Supplemental Information includes five figures, four tables, and three data files and can be found with this article online at http://dx.doi.org/10.1016/j.str.2017.05.003.

AUTHOR CONTRIBUTIONS

Conceptualization, M.R.R., J.L., and R.Z.G.; Methodology, R.Z.G., R.M.P., and E.H.D.; Investigation, R.M.P., E.H.D., W.G.W., Y.X., M.S.L., K.A.B., S.J.P., L.M.P., B.C.C., H.N.I., R.R.B., A.P.B., and R.Z.G.; Writing – Original Draft, M.R.R., R.M.P., and E.H.D.; Writing – Review & Editing, M.R.R., R.M.P., E.H.D., R.Z.G., J.L., Y.X., W.G.W., M.S.L., K.A.B., and S.J.P.; Funding Acquisition, M.R.R. and J.L.; Supervision, M.R.R., R.Z.G., and R.M.P.

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REFERENCES

Abuhelwa, A.Y., Foster, D.J.R., and Upton, R.N. (2016). A quantitative review and meta-models of the variability and factors affecting oral drug absorption—part I: gastrointestinal pH. AAPS J. *18*, 1309–1321.

Adams, P.D., Grosse-Kunstleve, R.W., and Hung, L.W. (2002). PHENIX: building new software for automated crystallographic structure determination. Acta Crystallogr. D Biol. Crystallogr. *58*, 1948–1954.

Altschul, S.F., Gish, W., Miller, W., and Myers, E.W. (1990). Basic local alignment search tool. J. Mol. Biol. *215*, 403–410.

Auclair, S.M., Bhanu, M.K., and Kendall, D.A. (2012). Signal peptidase I: cleaving the way to mature proteins. Protein Sci. *21*, 13–25.

Biasini, M., Bienert, S., Waterhouse, A., Arnold, K., Studer, G., Schmidt, T., Kiefer, F., Cassarino, T.G., Bertoni, M., Bordoli, L., et al. (2014). SWISS-MODEL: modelling protein tertiary and quaternary structure using evolutionary information. Nucleic Acids Res. *42*, W252–W258.

Boelsterli, U.A., Redinbo, M.R., and Saitta, K. (2012). Multiple NSAID-induced hits injure the small intestine: underlying mechanisms and novel strategies. Toxicol. Sci. *131*, 654–667.

Boraston, A.B., Bolam, D.N., Gilbert, H.J., and Davies, G.J. (2004). Carbohydrate-binding modules: fine-tuning polysaccharide recognition. Biochem. J. *382*, 769–781.

Callanan, M.J., Russell, W.M., and Klaenhammer, T.R. (2007). Modification of Lactobacillus beta-glucuronidase activity by random mutagenesis. Gene *389*, 122–127.

David, L.A., Maurice, C.F., Carmody, R.N., Gootenberg, D.B., Button, J.E., Wolfe, B.E., Ling, A.V., Devlin, A.S., Varma, Y., Fischbach, M.A., et al. (2013). Diet rapidly and reproducibly alters the human gut microbiome. Nature *505*, 559–563.

Emsley, P., and Cowtan, K. (2004). Coot: model-building tools for molecular graphics. Acta Crystallogr. D Biol. Crystallogr. *60*, 2126–2132.

Finn, R.D., Coggill, P., Eberhardt, R.Y., Eddy, S.R., Mistry, J., Mitchell, A.L., Potter, S.C., Punta, M., Qureshi, M., Sangrador-Vegas, A., et al. (2016). The Pfam protein families database: towards a more sustainable future. Nucleic Acids Res. *44*, D279–D285.

Gloux, K., Berteau, O., El Oumami, H., Béguet, F., Leclerc, M., and Doré, J. (2011). A metagenomic β-glucuronidase uncovers a core adaptive function of the human intestinal microbiome. Proc. Natl. Acad. Sci. USA *108* (*Suppl*), 4539–4546.

Hassan, M.I., Waheed, A., Grubb, J.H., Klei, H.E., Korolev, S., and Sly, W.S. (2013). High resolution crystal structure of human beta-glucuronidase reveals structural basis of lysosome targeting. PLoS One *8*, e79687.

Huson, D.H., and Scornavacca, C. (2012). Dendroscope 3: an interactive tool for rooted phylogenetic trees and networks. Syst. Biol. *61*, 1061–1067.

Juncker, A.S., Willenbrock, H., Von Heijne, G., Brunak, S., Nielsen, H., and Krogh, A. (2003). Prediction of lipoprotein signal peptides in Gram-negative bacteria. Protein Sci. *12*, 1652–1662.

Kabsch, W. (2010). XDS. Acta Crystallogr. Sect. D Biol. Crystallogr. *66*, 125–132.

Kau, A.L., Ahern, P.P., Griffin, N.W., Goodman, A.L., and Gordon, J.I. (2011). Human nutrition, the gut microbiome and the immune system. Nature *474*, 327–336.

Kelley, L.A., Mezulis, S., Yates, C.M., Wass, M.N., and Sternberg, M.J.E. (2015). The Phyre2 web portal for protein modeling, prediction and analysis. Nat. Protoc. *10*, 845–858.

Kim, D.H., Hong, S.W., Kim, B.T., Bae, E.A., Park, H.Y., and Han, M.J. (2000). Biotransformation of glycyrrhizin by human intestinal bacteria and its relation to biological activities. Arch. Pharm. Res. *23*, 172–177.

Koropatkin, N.M., Cameron, E.A., and Martens, E.C. (2012). How glycan metabolism shapes the human gut microbiota. Nat. Rev. Microbiol. *10*, 323–335.

Langmead, B., and Salzberg, S.L. (2012). Fast gapped-read alignment with Bowtie 2. Nat. Methods *9*, 357–359.

Liang, W.J., Wilson, K.J., Xie, H., Knol, J., Suzuki, S., Rutherford, N.G., Henderson, P.J.F., and Jefferson, R.A. (2005). The gusBC genes of *Escherichia coli* encode a glucuronide transport system. J. Bacteriol. *187*, 2377–2385.

Liao, Y., Smyth, G.K., and Shi, W. (2014). FeatureCounts: an efficient general purpose program for assigning sequence reads to genomic features. Bioinformatics *30*, 923–930.

LoGuidice, A., Wallace, B.D., Bendel, L., Redinbo, M.R., and Boelsterli, U.A. (2012). Pharmacologic targeting of bacterial β-glucuronidase alleviates nonsteroidal anti-inflammatory drug-induced enteropathy in mice. J. Pharmacol. Exp. Ther. *341*, 447–454.

Lombard, V., Ramulu, H.G., Drula, E., Coutinho, P.M., and Henrissat, B. (2014). The carbohydrate-active enzymes database (CAZy) in 2013. Nucleic Acids Res. *42*, D490–D495.

Mackenzie, P.I., Owens, I.S., Burchell, B., Bock, K.W., Bairoch, A., Bélanger, A., Fournel-Gigleux, S., Green, M., Hum, D.W., Iyanagi, T., et al. (1997). The UDP glycosyltransferase gene superfamily: recommended nomenclature update based on evolutionary divergence. Pharmacogenetics *7*, 255–269.

Mani, S., Boelsterli, U.A., and Redinbo, M.R. (2014). Interrogating and modulating mammalian-microbial communication for improved health. Annu. Rev. Pharmacol. Toxicol. *55*, 559–580.

Masamune, H. (1934). Biochemical studies on carbohydrates. IV. On an enzyme which catalyses the hydrolysis of biosynthetic osides of glucuronic acid. J. Biochem. *19*, 353–375.

Matsumura, I., and Ellington, A.D. (2001). In vitro evolution of beta-glucuronidase into a beta-galactosidase proceeds through non-specific intermediates. J. Mol. Biol. *305*, 331–339.

Maurice, C.F., Haiser, H.J., and Turnbaugh, P.J. (2013). Xenobiotics shape the physiology and gene expression of the active human gut microbiome. Cell *152*, 39–50.

McBain, A.J., and Macfarlane, G.T. (1998). Ecological and physiological studies on large intestinal bacteria in relation to production of hydrolytic and reductive enzymes involved in formation of genotoxic metabolites. J. Med. Microbiol. *47*, 407–416.

Meech, R., Miners, J.O., Lewis, B.C., and MacKenzie, P.I. (2012). The glycosidation of xenobiotics and endogenous compounds: versatility and redundancy in the UDP glycosyltransferase superfamily. Pharmacol. Ther. *134*, 200–218.

Methé, B.A., Nelson, K.E., Pop, M., Creasy, H.H., Giglio, M.G., Huttenhower, C., Gevers, D., Petrosino, J.F., Abubucker, S., Badger, J.H., et al. (2012). A framework for human microbiome research. Nature *486*, 215–221.

Human Microbiome Project Consortium. (2012). Structure, function and diversity of the healthy human microbiome. Nature *486*, 207–214.

Nakamura, J., Kubota, Y., Miyaoka, M., Saitoh, T., Mizuno, F., and Benno, Y. (2002). Comparison of four microbial enzymes in Clostridia and Bacteroides isolated from human feces. Microbiol. Immunol. *46*, 487–490.

Navarre, W.W., and Schneewind, O. (1999). Surface proteins of gram-positive bacteria and mechanisms of their targeting to the cell wall envelope. Microbiol. Mol. Biol. Rev. *63*, 174–229.

Oshima, G. (1934). Biochemical studies on carbohydrates. XII. On beta-glucuronidase, 2nd communication. J. Biochem. *20*, 361–370.

Petersen, T.N., Brunak, S., von Heijne, G., and Nielsen, H. (2011). SignalP 4.0: discriminating signal peptides from transmembrane regions. Nat. Methods *8*, 785–786.

Redinbo, M.R. (2014). The microbiota, chemical symbiosis, and human disease. J. Mol. Biol. *426*, 3877–3891.

Roberts, A.B., Wallace, B.D., Venkatesh, M.K., Mani, S., and Redinbo, M.R. (2013). Molecular insights into microbial β-glucuronidase inhibition to abrogate CPT-11 toxicity. Mol. Pharmacol. *84*, 208–217.

Russell, W.M., and Klaenhammer, T.R. (2001). Identification and cloning of gusA, encoding a new beta-glucuronidase from *Lactobacillus gasseri* ADH. Appl. Environ. Microbiol. *67*, 1253–1261.

Saitta, K.S., Zhang, C., Lee, K.K., Fujimoto, K., Redinbo, M.R., and Boelsterli, U.A. (2014). Bacterial beta-glucuronidase inhibition protects mice against enteropathy induced by indomethacin, ketoprofen, or diclofenac: mode of action and pharmacokinetics. Xenobiotica *44*, 28–35.

Salyers, A.A., and O'Brien, M. (1980). Cellular location of enzymes involved in chondroitin sulfate breakdown by *Bacteroides thetaiotaomicron*. J. Bacteriol. *143*, 772–780.

Sievers, F., Wilm, A., Dineen, D., Gibson, T.J., Karplus, K., Li, W., Lopez, R., McWilliam, H., Remmert, M., Söding, J., et al. (2011). Fast, scalable generation of high-quality protein multiple sequence alignments using Clustal Omega. Mol. Syst. Biol. *7*, 539.

Takasuna, K., Hagiwara, T., Hirohashi, M., Kato, M., Nomura, M., Nagai, E., Yokoi, T., and Kamataki, T. (1996). Involvement of beta-glucuronidase in intestinal microflora in the intestinal toxicity of the antitumor camptothecin derivative irinotecan hydrochloride (CPT-11) in rats. Cancer Res. *56*, 3752–3757.

Turnbaugh, P.J., Hamady, M., Yatsunenko, T., Cantarel, B.L., Duncan, A., Ley, R.E., Sogin, M.L., Jones, W.J., Roe, B.A., Affourtit, J.P., et al. (2008). A core gut microbiome in obese and lean twins. Nature *457*, 480–484.

Wallace, B.D., Wang, H., Lane, K.T., Scott, J.E., Orans, J., Koo, J.S., Venkatesh, M., Jobin, C., Yeh, L.-A., Mani, S., et al. (2010). Alleviating cancer drug toxicity by inhibiting a bacterial enzyme. Science *330*, 831–835.

Wallace, B.D., Roberts, A.B., Pollet, R.M., Ingle, J.D., Biernat, K.A., Pellock, S.J., Venkatesh, M.K., Guthrie, L., O'Neal, S.K., Robinson, S.J., et al. (2015). Structure and inhibition of microbiome β-glucuronidases essential to the alleviation of cancer drug toxicity. Chem. Biol. *22*, 1238–1249.

Wilson, K.J., Hughes, S.G., and Jefferson, R.A. (1992). The *Escherichia coli* gus operon: induction and expression of the gus operon in *E. coli* and the occurrence and use of GUS in other bacteria. In In GUS Protocols. Using the GUS Gene as Reporter of Gene Expression, S.R. Gallagher, ed. (Academic Press), pp. 7–22.

Xu, Y., Cai, C., Chandarajoti, K., Hsieh, P.H., Li, L., Pham, T.Q., Sparkenbaugh, E.M., Sheng, J., Key, N.S., Pawlinski, R., et al. (2014). Homogeneous low-molecular-weight heparins with reversible anticoagulant activity. Nat. Chem. Biol. *10*, 248–250.

Zückert, W.R. (2014). Secretion of bacterial lipoproteins: through the cytoplasmic membrane, the periplasm and beyond. Biochim. Biophys. Acta *1843*, 1509–1516.

STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Chemicals, Peptides, and Recombinant Proteins		
Lysogeny Broth (LB) media	Fisher	Product code: BP1426
Isopropyl-1-thio-D-galactopyranoside (IPTG)	Fisher	Product code: BP1755
Roche complete-EDTA free protease inhibitor tablets	Sigma	Product code: 05056489001
Para-nitrophenyl glucuronide (PNPG)	Sigma	Product code: N1627
PEG 3350	Sigma	Product code: P3640
heparan nonasaccharide substrate [GlcUA-(GlcNAc-GlcUA)$_4$-PNP]	Synthesized in-house	(Xu et al., 2014)
Critical Commercial Assays		
GE Ni HisTrap HP 5 ml Column	GE Healthcare Life Sciences	Product code: 17-5255-01
Hi Load 16/60 Superdex 200 pg Column	GE Healthcare Life Sciences	Product code: 28989335
Polyamine-based anion exchange (PAMN) column (0.46cm x 25 cm)	Waters Corporation	Product code: YMC PB12S05-2546WT
C18 Column	ThermoFisher	Product code: 72105-254630
Crystallization Trays	Qiagen	Product code: 132006
Deposited Data		
Coordinates of *B. uniformis* β-glucuronidase structure	This paper	5UJ6
Experimental Models: Cell Lines		
E. coli BL21 DE3 Gold	Agilent Technologies	Cat. #230132
Oligonucleotides		
Primers for β-glucuronidase enzyme cloning, see Table S4	This paper	IDT DNA
Recombinant DNA		
Eubacterium eligens β-glucuronidase (EeGUS) gene in pUC57 vector	Codon optimized for protein sequence WP_012740861	BIO BASIC INC.
Faecalibacterium prausnitzii β-glucuronidase (FpGUS) gene in pUC57 vector	Codon optimized for protein sequence WP_005931592	BIO BASIC INC.
Bacteroides uniformis β-glucuronidase (BuGUS) gene in pUC57 vector	Codon optimized for protein sequence WP_035447612	BIO BASIC INC.
Parabacteroides merdae β-glucuronidase (PmGUS) gene in pUC57 vector	Codon optimized for protein sequence WP_005639106	BIO BASIC INC.
Bacteroides ovatus β-glucuronidase (BoGUS) gene in pUC57 vector	Codon optimized for protein sequence WP_004298526	BIO BASIC INC.
Bacteroides dorei β-glucuronidase (BdGUS) gene in pUC57 vector	Codon optimized for protein sequence WP_007841259	BIO BASIC INC.
pET28a-*E. coli* β-glucuronidase	Lab stock	(Wallace et al., 2010)
pLIC-His vector	Lab stock	Sonek Lab, University of North Carolina at Chapel Hill
EeGUS, FpGUS, BfGUS, BuGUS, PmGUS, BoGUS, BdGUS gene in pLIC-His vector	This paper	N/A
Bacteroides fragils β-glucuronidase (BfGUS) in pSGX3	DNASU Plasmid Repository	ID#BfCD00534345
Software and Algorithms		
NCBI Protein BLAST	(Altschul et al., 1990)	blast.ncbi.nlm.nih.gov/Blast.cgi
Clustal Omega	(Sievers et al., 2011)	www.ebi.ac.uk/Tools/msa/clustalo/

(*Continued on next page*)

Continued

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Protein Homology/analogy Recognition Engine V 2.0 (Phyre2)	(Kelley et al., 2015)	www.sbg.bio.ic.ac.uk/phyre2/html/page.cgi
SWISS-MODEL	(Biasini et al., 2014)	https://swissmodel.expasy.org
Dendroscope version 3.5.7	(Huson and Scornavacca, 2012)	dendroscope.org
SignalP 4.1 Server	(Petersen et al., 2011)	http://www.cbs.dtu.dk/services/SignalP/
LipoP 1.0 Server	(Juncker et al., 2003)	http://www.cbs.dtu.dk/services/LipoP/
Bowtie2 (v.2.2.5)	(Langmead and Salzberg, 2012)	http://bowtie-bio.sourceforge.net/bowtie2/index.shtml
Featurecount from the subread package (v.1.4.6-p2-Linux-x86_64)	(Liao et al., 2014)	subread.sourceforge.net
X-ray Detector Software (XDS)	(Kabsch, 2010)	http://xds.mpimf-heidelberg.mpg.de
Python-based Hierarchical ENvironment for Integrated Xtallography (PHENIX)	(Adams et al., 2002)	www.phenix-online.org
Crystallographic Object-Oriented Tookit (COOT)	(Emsley and Cowtan, 2004)	http://www2.mrc-lmb.cam.ac.uk/personal/pemsley/coot
PyMOL	Molecular Graphics System, Schrodinger, LLC	www.pymol.org
Xcalibur 1.3	Thermo-Finnigan	www.thermoquest.com
Other		
Sequence data for HMGI3013 database	Human Microbiome Project	http://hmpdacc.org/HMGI
Sequence data for HMGC279 database	Human Microbiome Project	http://hmpdacc.org/HMGC
Microbial assemblies, annotations and reads for HMGI and HMGC stool samples	Human Microbiome Project	http://hmpdacc.org/HMASM
Previously characterized EcGUS sequence	NCBI	Accession: NP_416134.1
Previously characterized CpGUS sequence	NCBI	Accession: WP_003467686.1
Previously characterized SaGUS sequence	NCBI	Accession: WP_000966715.1
Previously characterized BfGUS sequence	NCBI	Accession: 3CMG
Homo sapiens β-glucuronidase sequence	NCBI	Accession: NP_000172.2
H11G11-BG from uncultured bacterium	NCBI	Accession: CBJ55484.1

CONTACT FOR REAGENT AND RESOURCE SHARING

Further information and requests for resources and reagents should be directed to and will be fulfilled by the Lead Contact, Matthew R Redinbo (redinbo@unc.edu).

EXPERIMENTAL MODEL AND SUBJECT DETAILS

Cell Lines

E. coli BL21 DE3 Gold cells were cultured in lysogeny broth (LB) with vigorous shaking at 37°C.

METHOD DETAILS

Human Microbiome Project (HMP) Data

Protein sequences for stool samples from the HMP Clustered genes- HMGC (http://hmpdacc.org/HMGC) and Gene Indices – HMGI (http://hmpdacc.org/HMGI/) were downloaded as FASTA files and used as a database for subsequent NCBI BLASTp searches (Altschul et al., 1990). Microbial assemblies, annotations and reads for stool samples were obtained from http://hmpdacc.org/HMASM/ and used for abundance calculations as described below.

HMP β-Glucuronidase Identification

In order to identify β-glucuronidase (GUS) enzymes in the HMP stool proteins, we used *Escherichia coli* β-glucuronidase (NCBI Accession: NP_416134.1), *Clostridium perfringens* β-glucuronidase (NCBI Accession: WP_003467686.1), *Streptococcus agalactiae*

β-glucuronidase (NCBI Accession: WP_000966715.1) and *Bacteroides fragilis* β-glucuronidase (NCBI Accession: 3CMG) in a pairwise alignment approach. Each HMP stool protein was aligned to the above proteins using NCBI BLASTp (Altschul et al., 1990). We considered alignments with an E-value < 0.05 and percent identity ≥25 to be valid hits. Only those valid hits were then checked for the presence of NxKG motif, catalytic E residues and N and Y motifs corresponding to GUS positions shown below.

Bacterium	NxKG Motif Positions	Catalytic E Residues Positions	N and Y Motifs Positions
Escherichia coli	566, 568, 569	413, 504	412, 468
Clostridium perfringens	567, 569, 570	412, 505	411, 468,
Streptococcus agalactiae	563, 565, 566	408, 501	407, 464
Bacteroides fragilis	547, 549, 550	395, 476	394, 445

We classified HMP proteins that satisfied the above conditions as β-glucuronidase.

HMP β-Glucuronidase Loop Classification

HMP GUS proteins identified above were subjected to a multiple sequence alignment (MSA) using Clustal Omega (Sievers et al., 2011) along with selected model GUS sequences: *E. coli* (NCBI Accession: NP_416134.1), *Streptococcus agalactiae* (NCBI Accession: WP_000966715.1), *Clostridium perfringens* (NCBI Accession: WP_003467686.1), *Homo sapiens* (NCBI Accession: NP_000172.2), H11G11-BG from uncultured bacterium (NCBI Accession: CBJ55484.1) and *Bacteroides fragilis* (NCBI Accession: 3CMG). The MSA was examined for the presence of two loop regions. The first (Loop 1) corresponds to *E. coli* GUS positions 356 to 380 and the second (Loop 2) corresponds to *E. coli* GUS positions 416 to 419. HMP GUS proteins were then categorized into six categories based on the number of residues present in those two regions after excluding gap characters as shown below.

Category	Residues in Loop 1 region	Residues in Loop 2 region
No Loop	< 10	< 9
Loop 1	> 15	< 9
Mini-Loop 1	≤ 15 and ≥ 10	< 9
Loop 2	< 10	≥ 12
Mini-Loop 2	< 10	≥ 9 and < 12
Mini-Loop 1,2	≤ 15 and ≥ 10	≥ 9 and < 12

Protein Structure Models

Protein structure models for the *P. merdae, B. ovatus, and B. dorei* GUS enzymes were generated using the Protein Homology/analogY Recognition Engine V 2.0 (Phyre2) (Kelley et al., 2015). The model for *B. uniformis* GUS was generated using SWISS-MODEL (Biasini et al., 2014). Models were analyzed and figures were generated in Pymol.

Taxonomic Assignments of GUSome Sequences

The non-redundant protein sequences database was searched for sequences matching protein sequences from the HMGC279 or HMGI3013 datasets using NCBI BLASTp (Altschul et al., 1990). For matches with ≥95% identity, phylum, genus and species taxonomy were assigned to that sequence in the HMGC279 or HMGI3013 dataset and any large sequencing errors such as duplication of a sequence segment, miscalled start or end sites were corrected. For matches with 25-94% sequence identity, the top results were analyzed for phylum of the expressing bacteria. If all results were from the same phylum, the HMGC279 or HMGI3013 sequence was assigned to that phylum. However, if sequences with ≥25% identity were found in multiple phyla, a phylum was not assigned and is instead noted as unclear. Potential sequencing errors were not corrected for these lower identity matches but were noted and a mature sequence length is not reported for these unverified sequences. Correction of sequencing errors in the HMGC279 dataset led to reduction of total sequences from 293 to 279 due to sequences that shared ≥95% sequence identity after correction. Similarly, sequences from the HMGI3013 dataset were considered to be equivalent sequences if they shared ≥95% sequence identity for the summaries presented in Figure 5. The radial cladogram in Figure 3A was generated in Dendroscope version 3.5.7 (Huson and Scornavacca, 2012). Data provided for HMGC279 and HMGI3013 sequences are based on the NCBI Protein Database as of August 2016 and October 2016, respectively.

Signal Peptide Identification

Following any correction of sequencing errors such as miscalled start sites found during analysis with BLASTp for taxonomic assignments, HMGC279 sequences were analyzed for signal peptide cleavage sites using the online SignalP 4.1 server for Gram-negative

bacteria (Petersen et al., 2011) and the online LipoP 1.0 server (Juncker et al., 2003). Sequences trimmed of the predicted signal sequence were manually interrogated for an N-terminal cysteine for possible lipidation for presentation on the outer membrane.

CAZy Enzyme and Module Family Identification

HMGC279 sequences were compared to the Carbohydrate-Active enZyme (CAZy) database in order to assign enzyme class and module family matches using the "annotate your proteins" function in the DataBase for automated Carbohydrate-active enzyme ANnotation (dbCAN) web server (dbCAN HMMs 5.0 based on CAZyDB 07/15/2016).

Gene Abundance Calculations

The trimmed and duplicate-marked reads for each stool sample were aligned to their own assembled gene sequences using bowtie2 (v.2.2.5) (Langmead and Salzberg, 2012). Alignments were stored in bam files and gene specific counts were obtained by running the program featurecount from the subread package (v.1.4.6-p2-Linux-x86_64) on each bam file (Liao et al., 2014). To account for differences in read numbers between samples, we normalized and log transformed the counts as shown below:

$$log_{10}\left(\frac{RC}{n} \times \frac{\sum x}{N} + 1\right)$$

where RC is the read count for a gene in a particular sample, n is the total number of reads in that sample, the sum of x is the total number of reads in all samples and N is the total number of samples.

Enzyme Cloning

Eubacterium eligens GUS

The codon-optimized *E. eligens* β-glucuronidase (*Ee*GUS) gene was purchased from Bio Basic in the pUC57 vector. The gene was amplified and inserted into the pLIC-His vector using the primers in Table S4, Primer Sequences, Related to Figure 5.

Faecalibacterium prausnitzii GUS

The *F. prausnitizii* β-glucuronidase (*Fp*GUS) gene was purchased from Bio Basic in the pUC57 vector. The gene was amplified and inserted into the pLIC- His vector using the primers in Table S4, Primer Sequences, Related to Figure 5.

Bacteroides fragilis

The mature gene lacking the signal peptide of *B. fragilis* β-glucuronidase (*Bf*GUS) was purchased from DNA SU Plasmid Repository in the pSGX3 plasmid. The gene was amplified and inserted into the pLIC-His vector using the primers in Table S4, Primer Sequences, Related to Figure 5.

Bacteroides uniformis GUS

The full-length *B. uniformis* β-glucuronidase (*Bu*GUS) gene was purchased from Bio Basic in the pUC57 vector. The mature gene lacking the signal peptide was amplified and inserted into the pLIC-His vector using the primers in Table S4, Primer Sequences, Related to Figure 5.

Parabacteroides merdae GUS

The full-length *P. merdae* β-glucuronidase (*Pm*GUS) gene was purchased from Bio Basic in the pUC57 vector. The mature gene lacking the signal peptide was amplified and inserted into the pLIC-His vector using the primers in Table S4, Primer Sequences, Related to Figure 5.

Bacteroides ovatus GUS

The full-length *B. ovatus* β-glucuronidase (*Bo*GUS) gene was purchased from Bio Basic in the pUC57 vector. The mature gene lacking the signal peptide was amplified and inserted into the pLIC-His vector using the primers in Table S4, Primer Sequences, Related to Figure 5.

Bacteroides dorei GUS

The full-length *B. dorei* β-glucuronidase (*Bd*GUS) gene was purchased from Bio Basic in the pUC57 vector. The mature gene lacking the signal peptide was amplified and inserted into the pLIC-His vector using the primers in Table S4, Primer Sequences, Related to Figure 5.

Expression and Purification of Enzymes

Each β-glucuronidase expression plasmid was transformed into BL21 DE3 Gold cells for enzyme expression. Cells were grown in the presence of kanamycin (EcGUS) or ampicillin (all other GUS constructs) in LB medium with vigorous shaking at 37°C to an OD600nm of 0.5, at which point the temperature was reduced to 18°C. At OD600=0.8, protein expression was induced by the addition of 0.1 mM isopropyl-1-thio-D-galactopyranoside (IPTG) and incubation continued overnight. For PmGUS and BoGUS, protein expression was carried out in auto-inducing media with shaking at 37°C until an OD600nm of ~1.0 was reached, at which point the temperature was reduced to 18°C. Cells were collected by centrifugation at 4500xg for 20 min at 4°C in a Sorvall (model RC-3B) swinging bucket centrifuge. Cell pellets were resuspended in Buffer A (20 mM Potassium Phosphate pH 7.4, 50 mM imidazole, 500 mM NaCl), DNase, lysozyme, and a Roche complete-EDTA free protease inhibitor tablet. Resuspended cells were sonicated and clarified via centrifugation at 17,000xg for 60 min in a Sorvall (model RC-5B). The lysate was flowed over a Ni-NTA HP column (GE Healthcare) loaded onto the Aktaxpress FPLC system (Amersham Bioscience) and washed with Buffer A. Protein was eluted with Buffer B (20 mM Potassium Phosphate pH 7.4, 250 mM Imidazole, 500mM NaCl). Fractions containing the protein of interest were combined and

passed over a HiLoadTM 16/60 SuperdexTM 200 gel filtration column. Protein was eluted in S200 Buffer (20 mM HEPES pH 7.4, 50 mM NaCl), except PmGUS which was eluted in S200 Buffer (20mM HEPES, pH 7.4, 300mM NaCl). Fractions were analyzed by SDS- PAGE and those with >95% purity were combined and concentrated for long-term storage at -80°C.

PNPG Assay

Para-nitrophenyl glucuronide (PNPG) was purchased as a solid and suspended in water to a concentration of 100 mM. Reactions were conducted in triplicate in 96-well, black, clear-bottom assay plates (Costar, Tewksbury MA) at 37°C. The reactions consisted of 10 μL assay buffer (50 mM HEPES, 50 mM NaCl, various pH), 10 μL enzyme, and 30 μL of substrate diluted in assay buffer. For each time point, a reaction was quenched with 100μL of 0.2 M sodium carbonate. When all reactions were quenched, product formation was measured via absorbance at 410 nm using a PHERAstar *Plus* microplate reader (BMG Labtech, Ortenberg, Germany). For analysis of the optimal pH for each enzyme, the above assay was run at one enzyme concentration (10 nM) and one PNPG concentration (800 μM) with buffer at pH 4, 4.5, 5, 5.5, 6, 6.5, 7, and 7.4. For kinetic assays, each protein was examined at its optimal pH using assay buffer and 800 μM PNPG diluted in the appropriate assay buffer. For assays at pH 6.5 to 7.4, initial 50 μL reactions described above were not quenched and product formation was measured over time via absorbance at 410 nm. For assays at pH 6 or lower, reactions were quenched and product formation in the final 150 μL reaction was measured.

Heparan Nonasaccharide Assay

The heparan nonasaccharide substrate [GlcUA-(GlcNAc-GlcUA)$_4$-PNP (where GlcUA is glucuronic acid, GlcNAc is N-acetylglucosamine)] was synthesized in-house (Xu et al., 2014) and was digested with the each of the eight GUS enzymes in the in vitro panel for 24 hours. Digestion reactions were composed of 0.5 μM GUS enzyme and 273 μM nonasaccharide. Aliquots of the resultant solutions were analyzed by polyamine-based anion exchange (PAMN)-HPLC. Sugars were eluted from the PAMN column (0.46 cm × 25 cm from Waters) with a linear gradient of KH_2PO_4 from 0 to 1 M in 40 min at a flow rate of 0.5 ml/ min. The eluent was monitored by a UV detector at 310 nm. Aliquots of heparan nonasaccharide digestion reactions were analyzed by electrospray ionization mass spectrometry (ESI-MS) by first purifying the reaction mixture by C_{18} column eluted with a linear gradient of methanol with 1% trifluoroacetic acid from 0 to 1 M in 60 min at a flow rate of 0.5 ml/ min. The purified oligosaccharides were then dried. ESI-MS analysis was performed on a Thermo LCQ-Deca in negative ionization mode. A syringe pump (Harvard Apparatus) was used to introduce the sample by direct infusion (50 μl /min). The purified oligosaccharides were diluted in 200 μl of H_2O with the electrospray source set to 3 KV and 150°C. The automatic gain control was set to 1×10^7 for full scan MS. The MS data were acquired and processed using Xcalibur 1.3.

Structure Determination of *B. uniformis* GUS

B. uniformis β-glucuronidase (BuGUS) was crystallized using hanging drop vapor diffusion at 20°C using a well solution consisting of 20% PEG 3350 and 0.2 M Potassium Chloride. 0.001 mL of well solution was combined with 0.001 mL of BuGUS (12 mg/mL in 20mM Hepes, pH 7.4, 100mM NaCl, 5% Glycerol and 0.5mM TCEP) and crystals appeared in 3 days. Crystal specimen were cryo-protected in 20% PEG 3350, 0.2 M Potassium Chloride and 15% glycerol and diffraction data were collected from one crystal at 100 °K to 1.9 Å resolution at APS Beamline 23-ID-B. The data were processed with XDS and the structure was determined by molecular replacement in Phenix (Adams et al., 2002) using the *B. fragilis* GUS structure (3CMG) as a search model. The structure was iteratively built by hand using COOT (Emsley and Cowtan, 2004) along with appropriate cycles of refinement using Phenix to final R and R_{free} values of 0.14 and 0.18, respectively (Table S3, BuGUS Crystallographic Analysis, Related to Figure 5). Coordinates and structure factors have been deposited at the RCSB with accession code 5UJ6.

QUANTIFICATION AND STATISTICAL ANALYSIS

PNPG Cleavage Rate and Heparan Cleavage in Figure 5A are reported as average and standard error of three reactions.

DATA AND SOFTWARE AVAILABILITY

The accession number for the coordinates and structure factors of the *B. uniformis* β-glucuronidase structure reported in this paper is PDB: 5UJ6. All other data are provided in supplemental files associated with this paper.

Structure, Volume *25*

Supplemental Information

An Atlas of β-Glucuronidases

in the Human Intestinal Microbiome

Rebecca M. Pollet, Emma H. D'Agostino, William G. Walton, Yongmei Xu, Michael S. Little, Kristen A. Biernat, Samuel J. Pellock, Loraine M. Patterson, Benjamin C. Creekmore, Hanna N. Isenberg, Rohini R. Bahethi, Aadra P. Bhatt, Jian Liu, Raad Z. Gharaibeh, and Matthew R. Redinbo

Figure S1A. Loop Category Variability per Individual. Related to Table 1, Table 2.

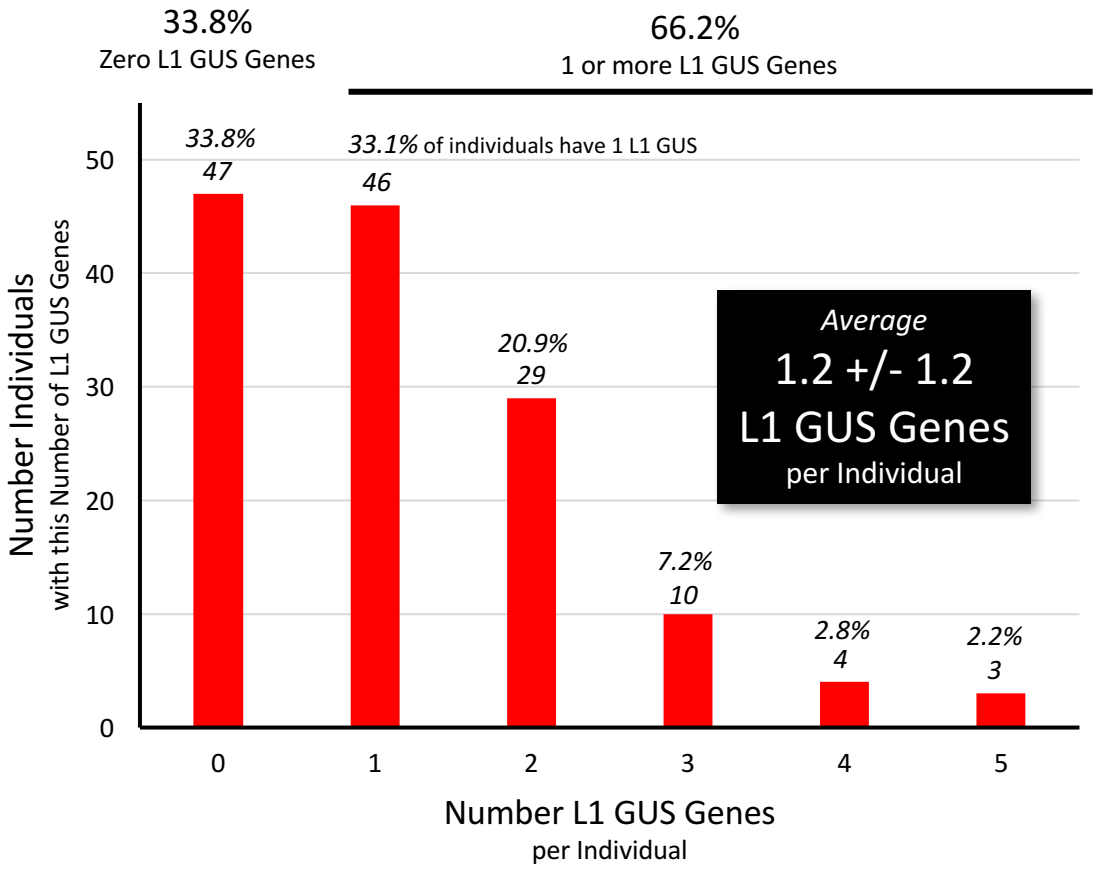


Figure S1B. Loop Category Variability per Individual. Related to Table 1, Table 2.

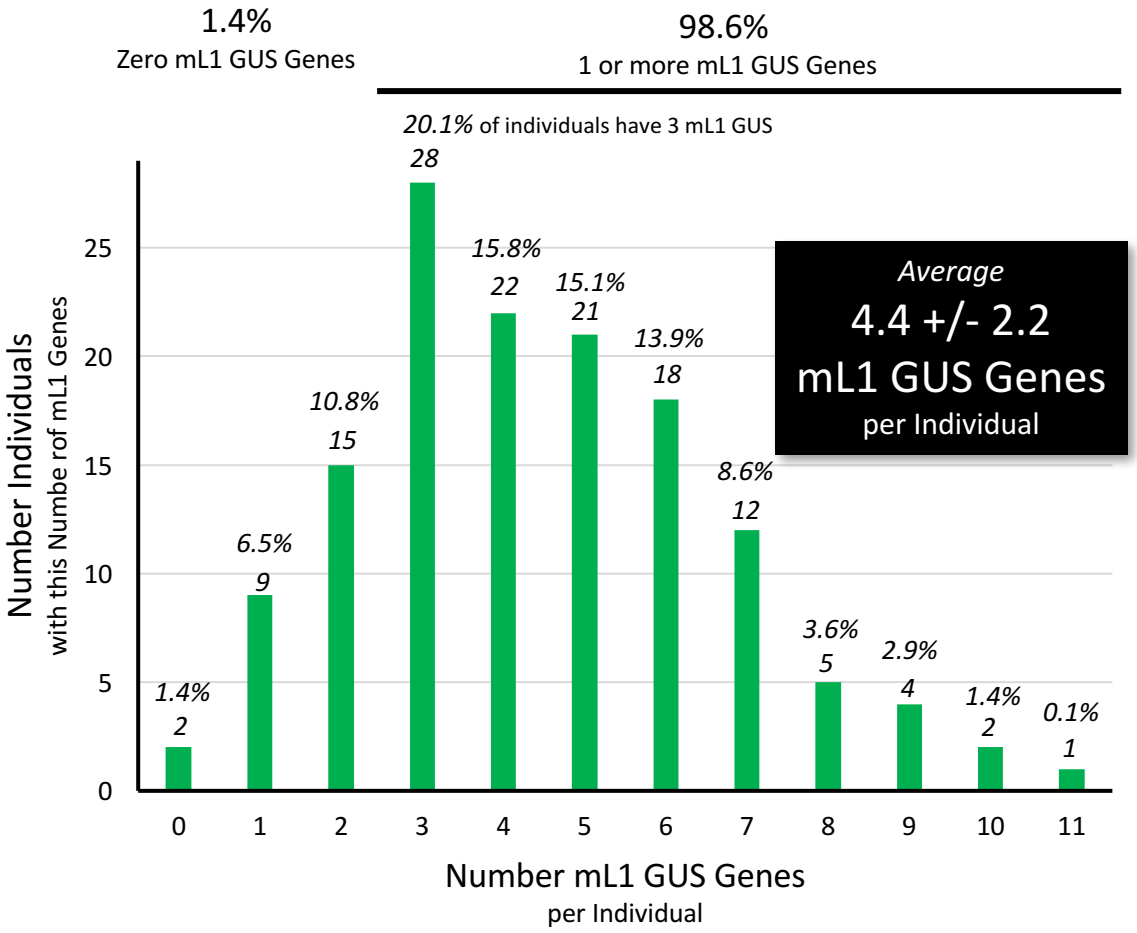


Figure S1C. Loop Category Variability per Individual. Related to Table 1, Table 2.

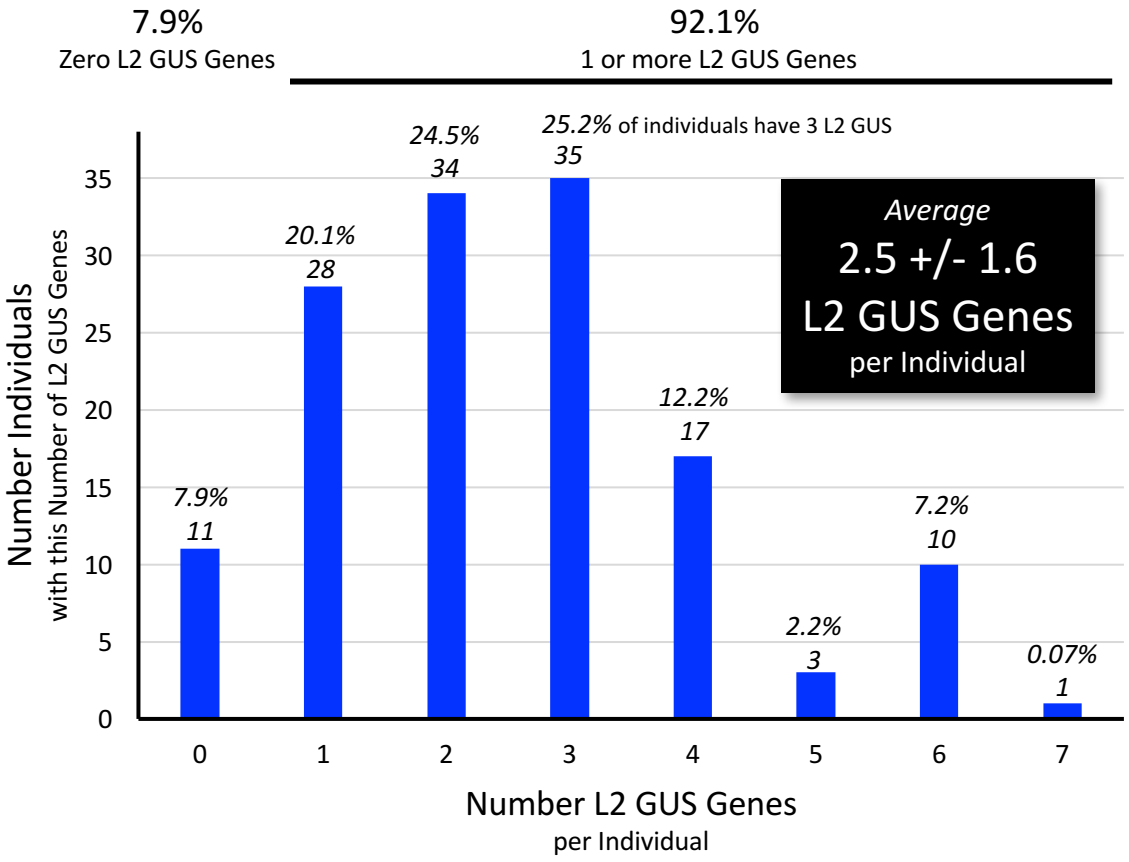


Figure S1D. Loop Category Variability per Individual. Related to Table 1, Table 2.

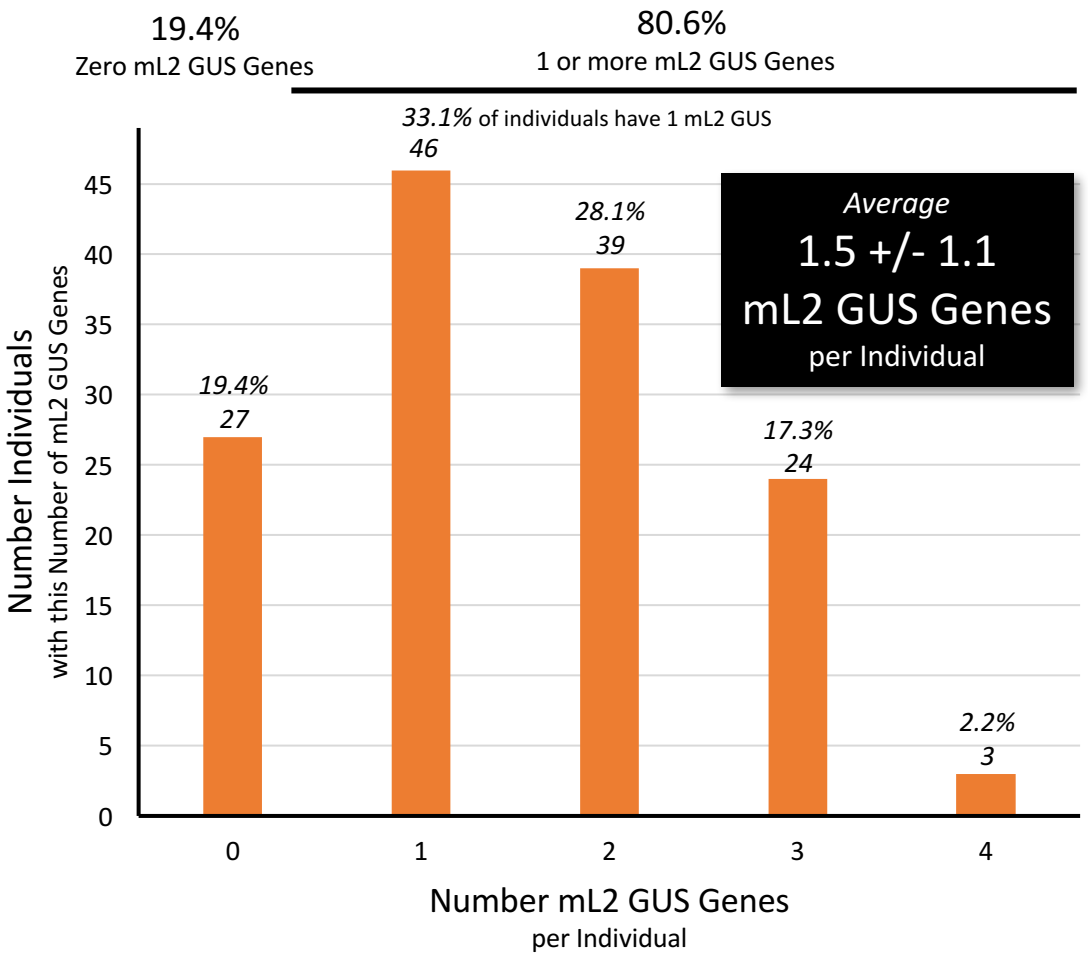


Figure S1E. Loop Category Variability per Individual. Related to Table 1, Table 2.

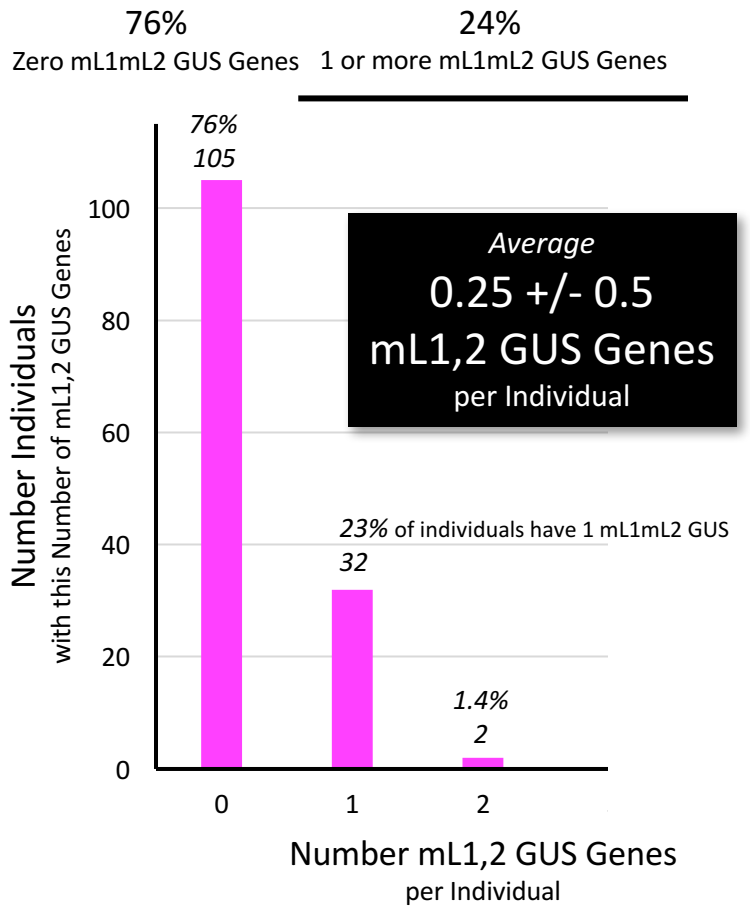


Figure S1F. Loop Category Variability per Individual. Related to Table 1, Table 2.

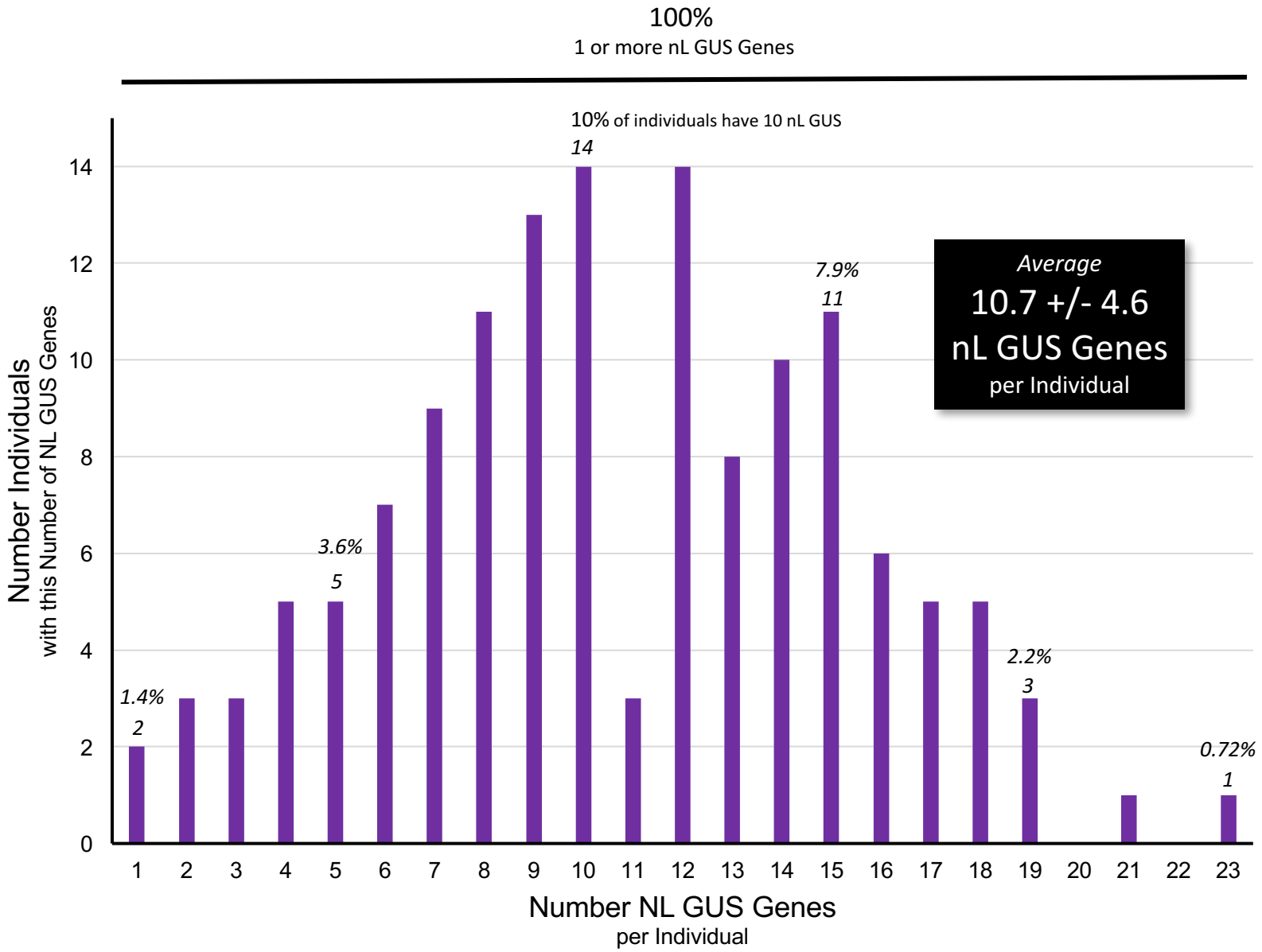


Figure S1G. Loop Category Variability per Individual. Related to Table 1, Table 2.

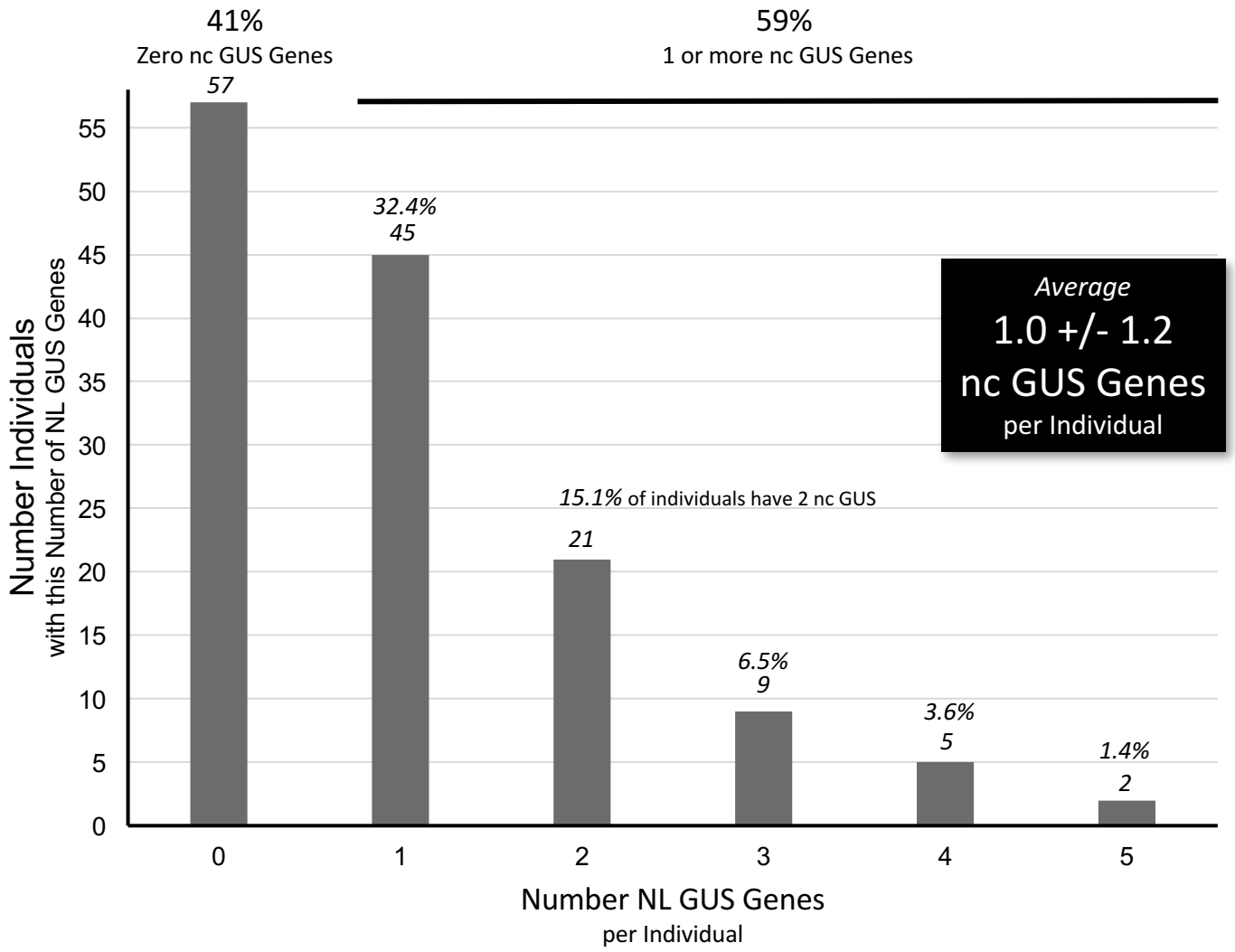


Figure S2. Summary of Loop Variability. Related to Table 2.

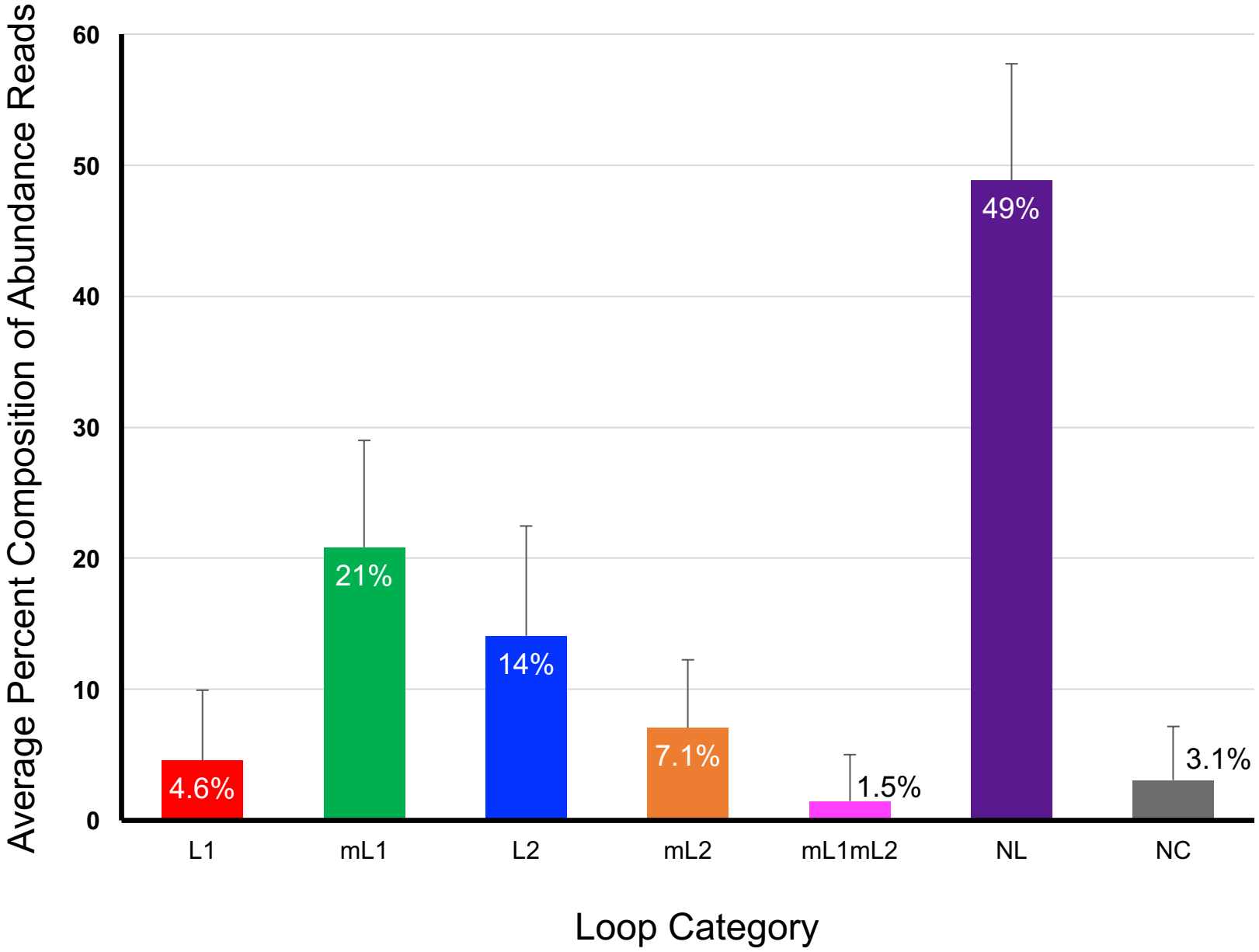


Figure S3. GUS Panel Sequence Identities. Related to Figure 5.

Pairwise Sequence Identities for GUS Panel Proteins

	Ec	Ee	Fp	Bf	Bu	Pm	Bo	Bd
Ec		45	45	26	26	25	25	26
Ee			40	26	25	21	26	26
Fp				27	26	24	26	25
Bf					34	28	42	31
Bu						29	30	35
Pm							27	31
Bo								31
Bd								

Figure S4A. GUS Panel pH Optima. Related to Figure 5.

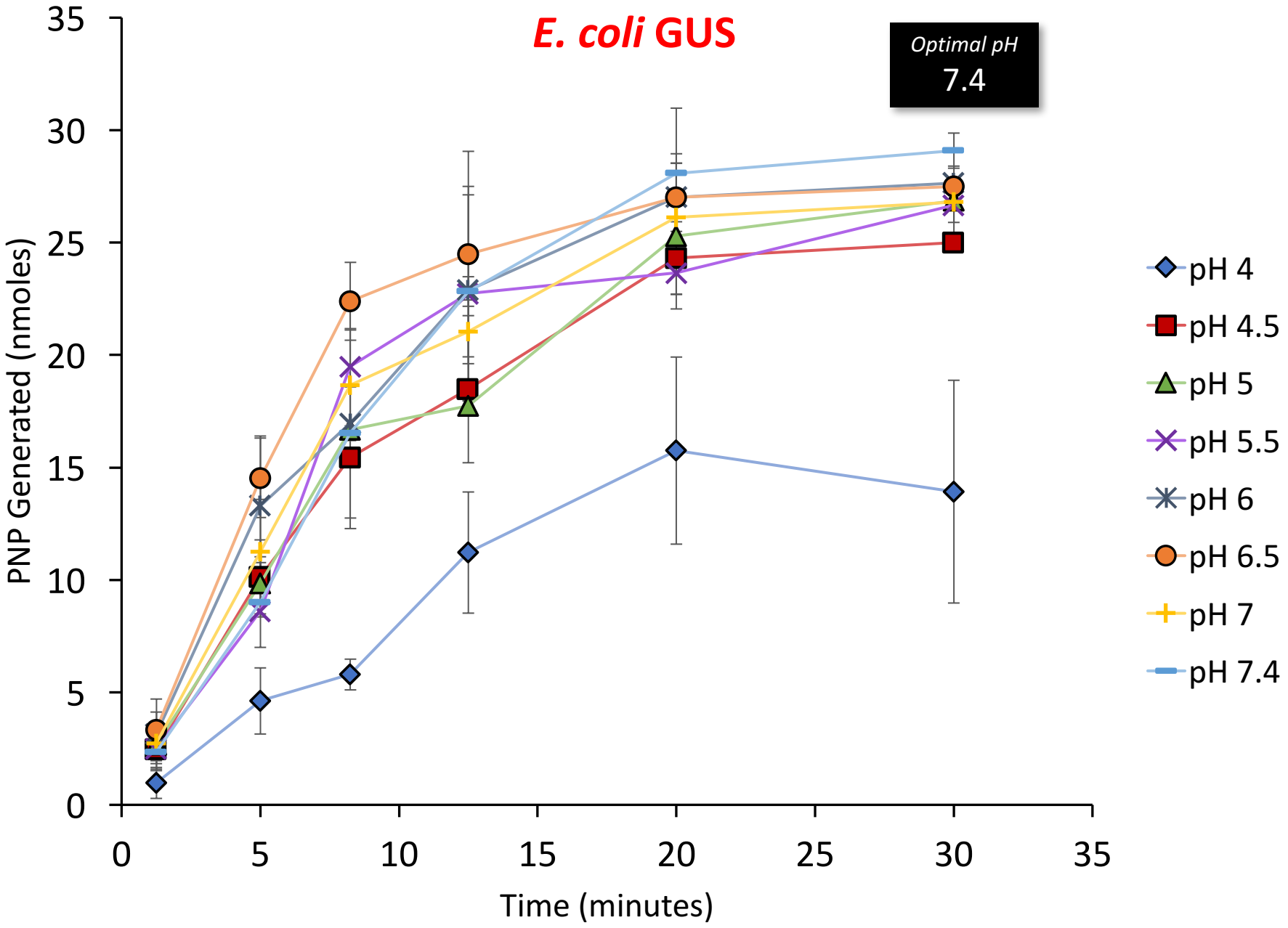


Figure S4B. GUS Panel pH Optima. Related to Figure 5.

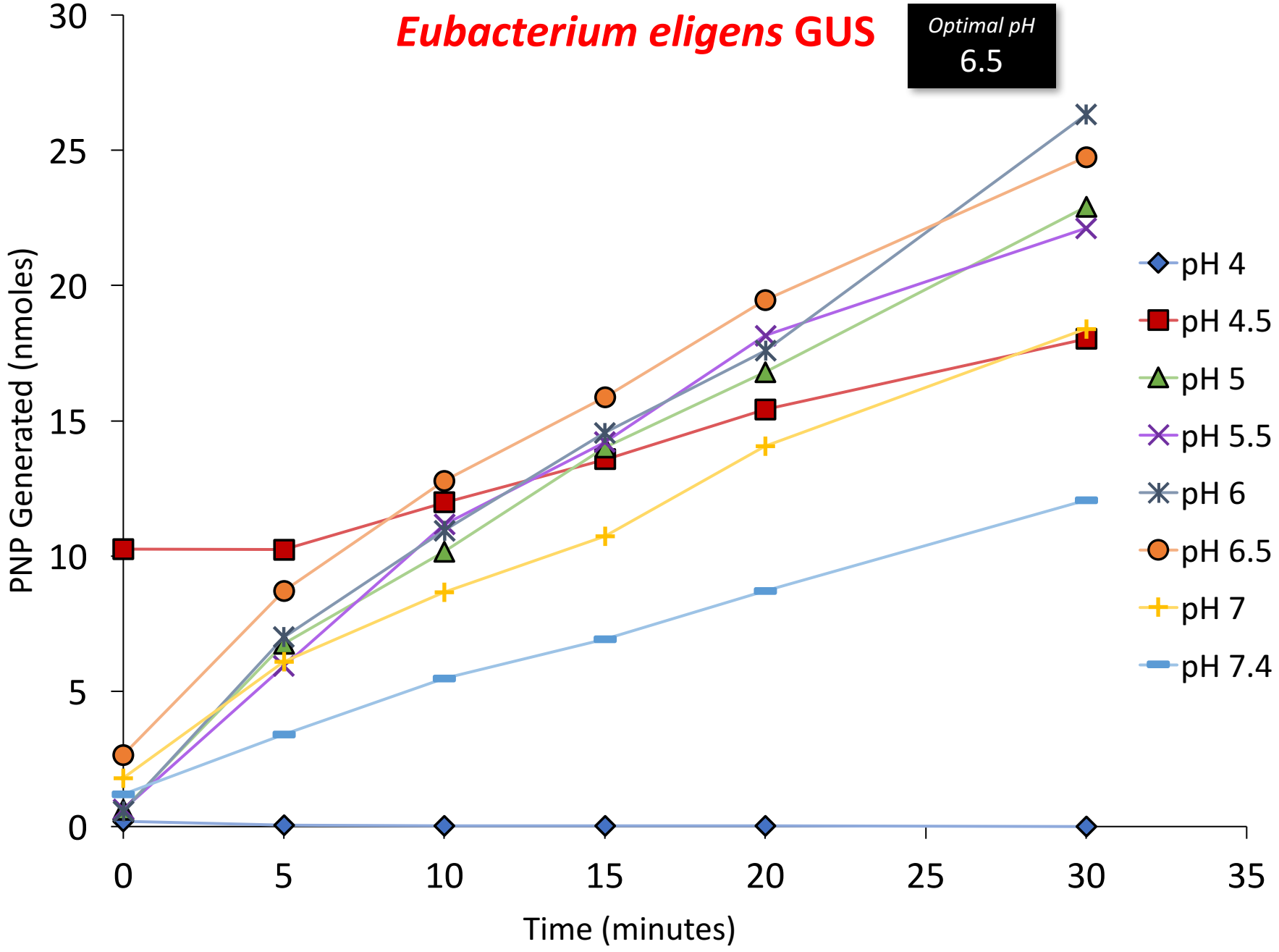


Figure S4C. GUS Panel pH Optima. Related to Figure 5.

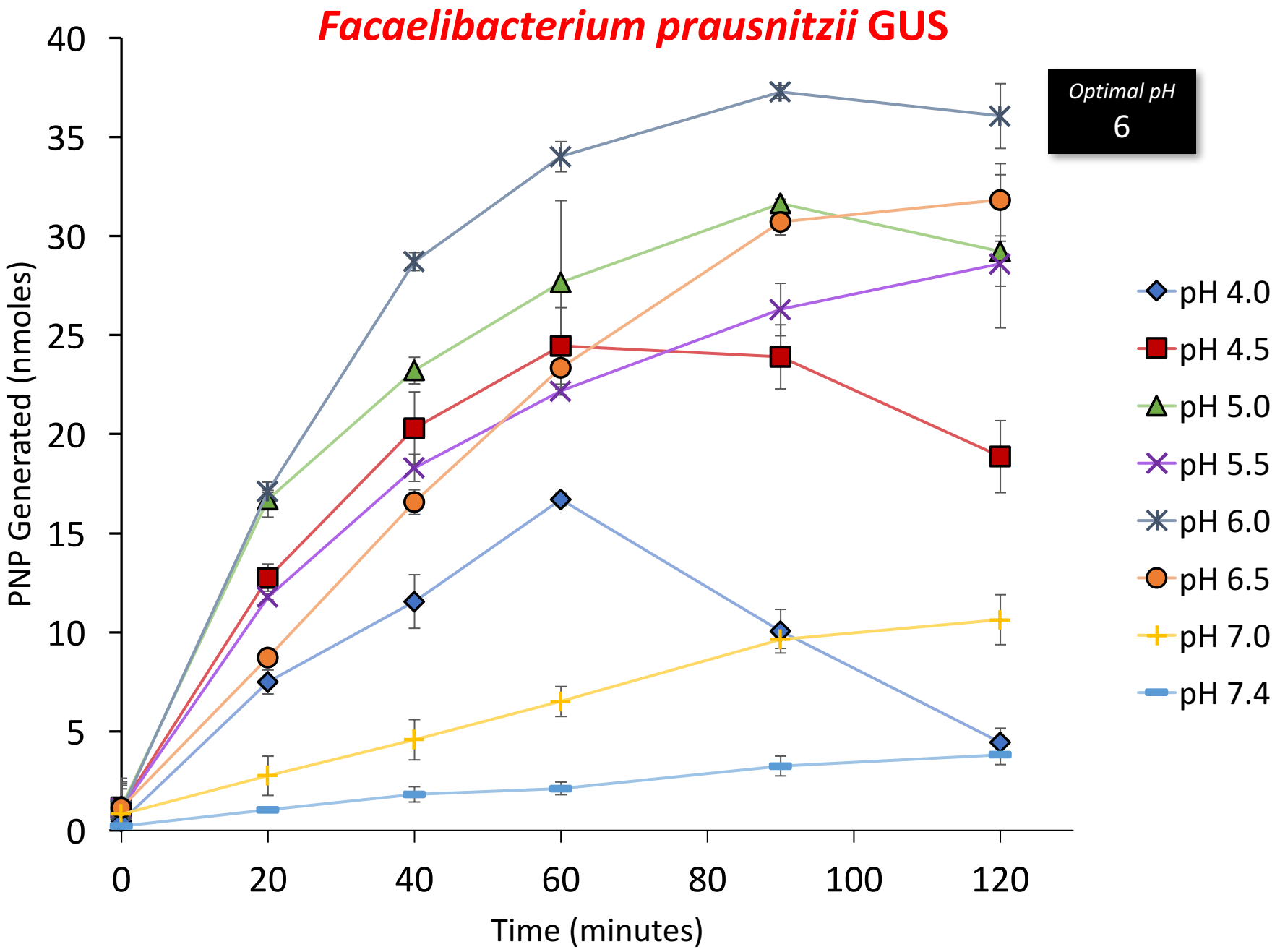


Figure S4D. GUS Panel pH Optima. Related to Figure 5.

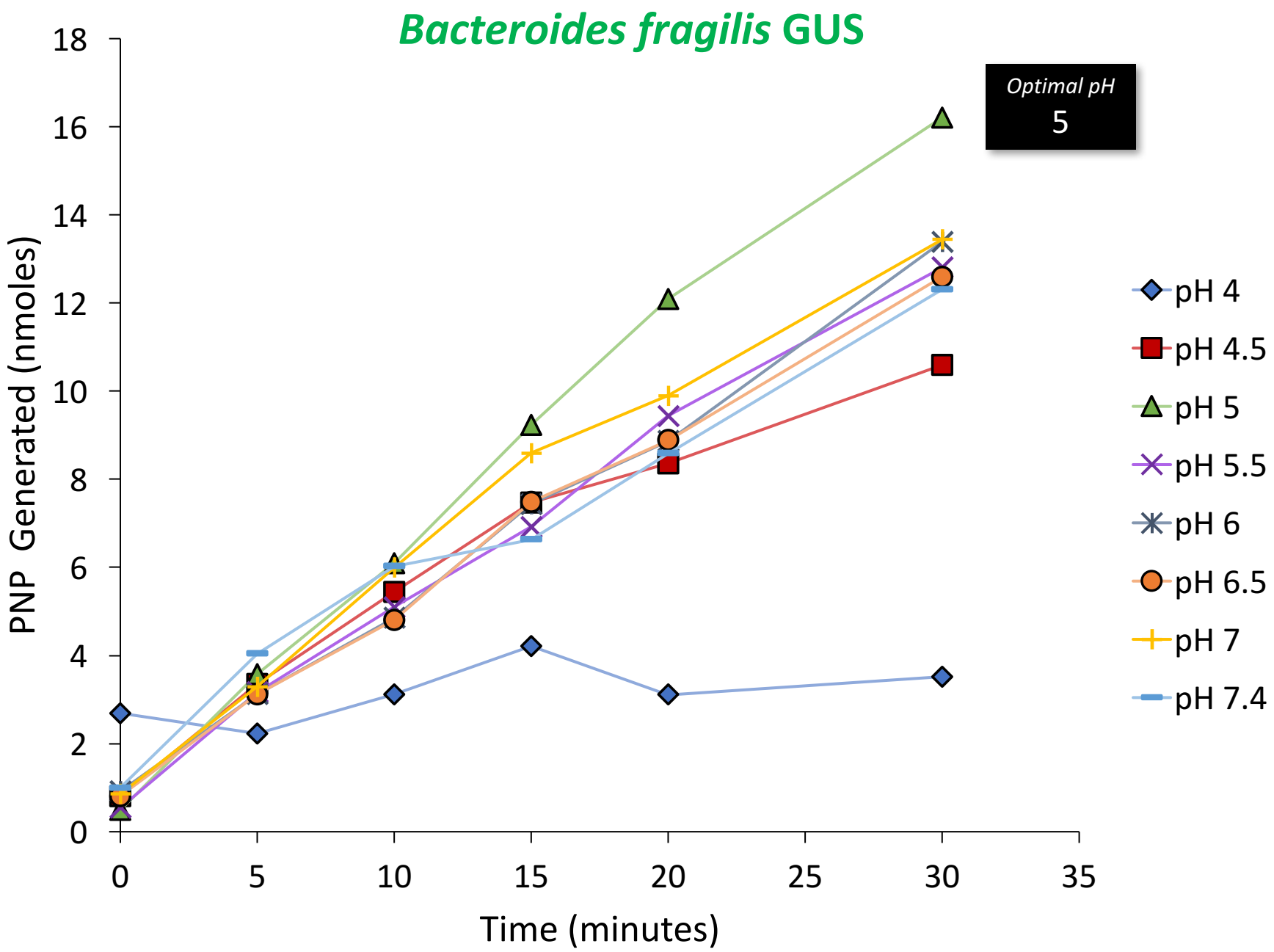


Figure S4E. GUS Panel pH Optima. Related to Figure 5.

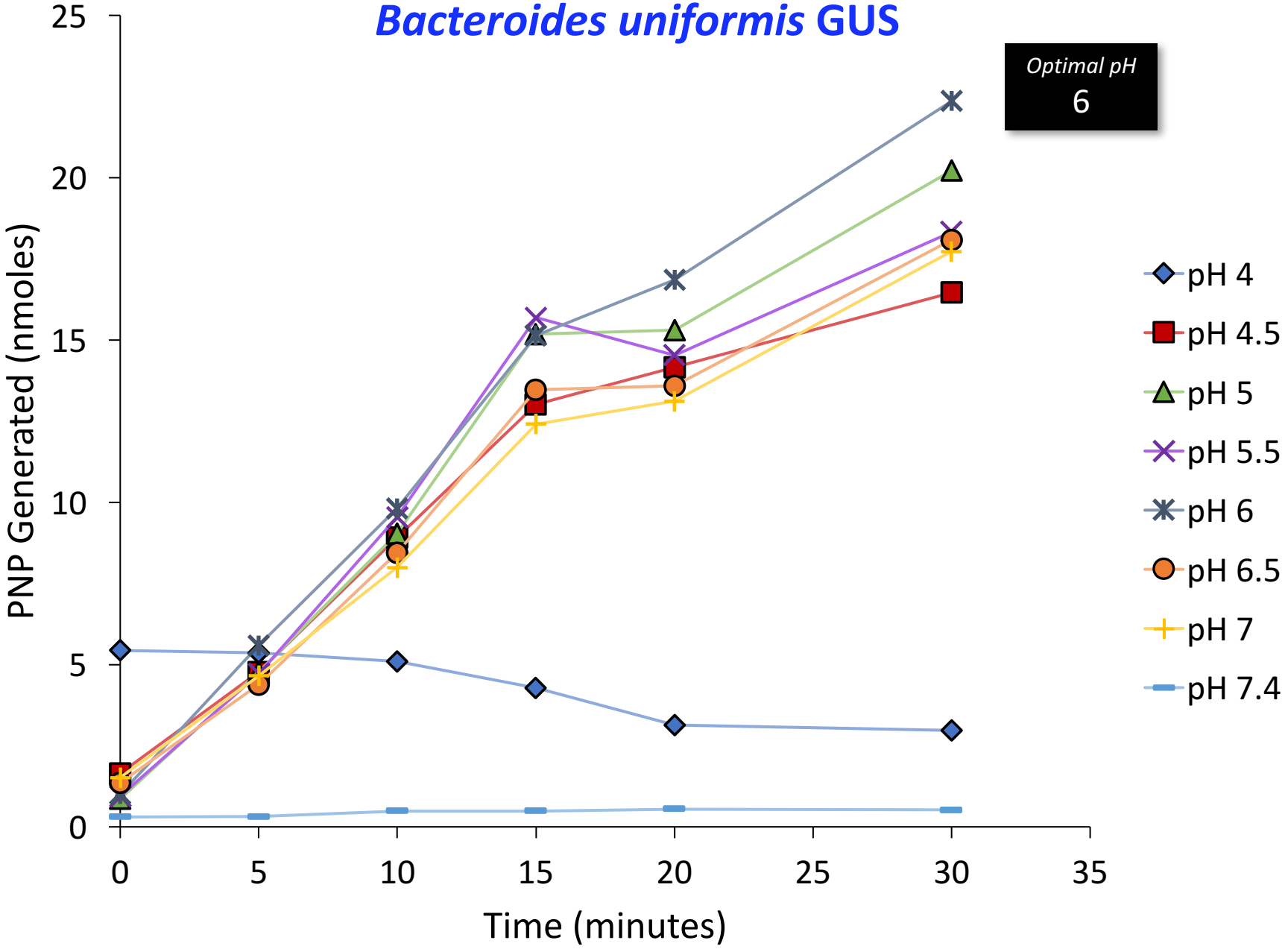


Figure S4F. GUS Panel pH Optima. Related to Figure 5.

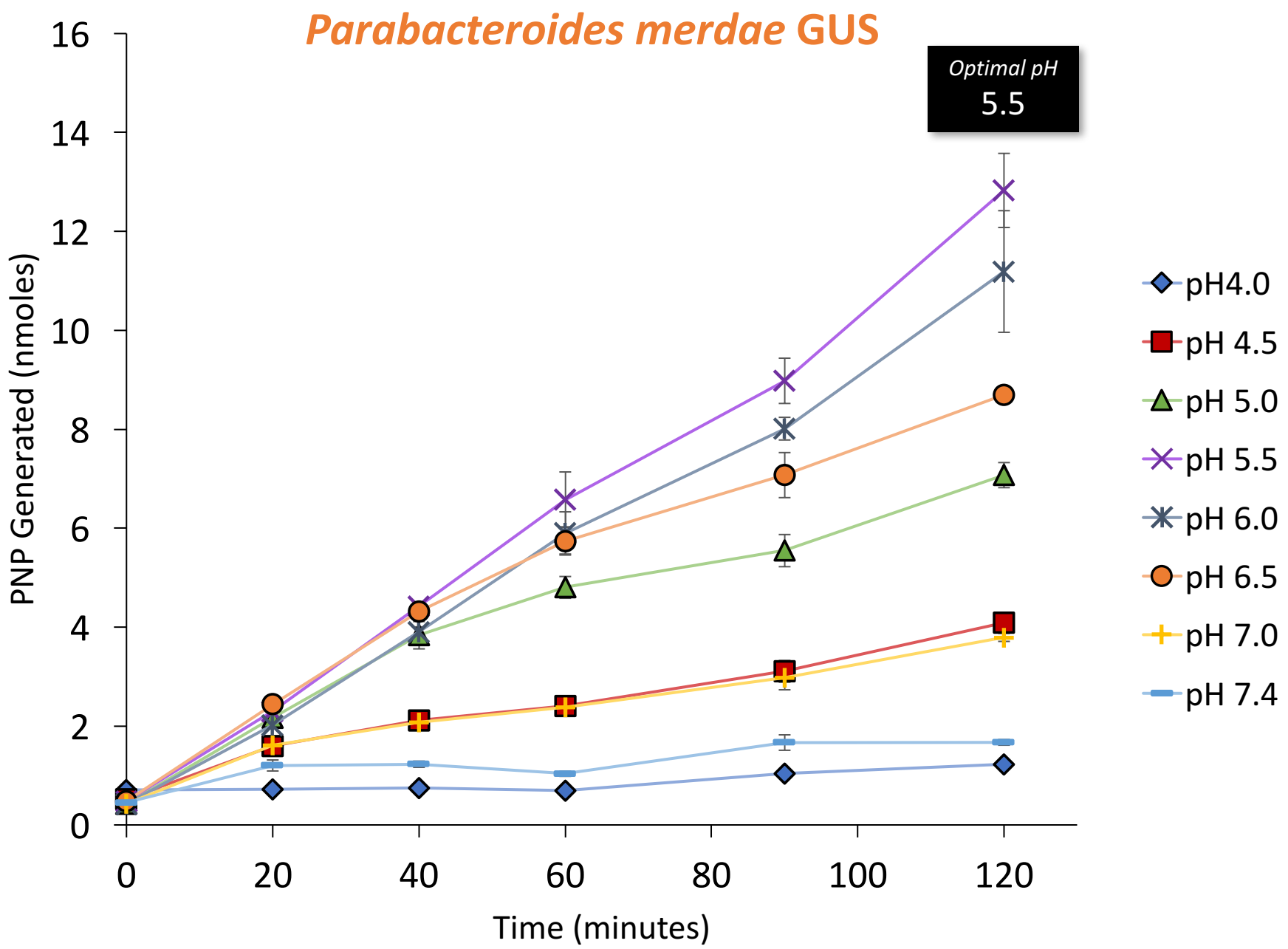


Figure S4G. GUS Panel pH Optima. Related to Figure 5.

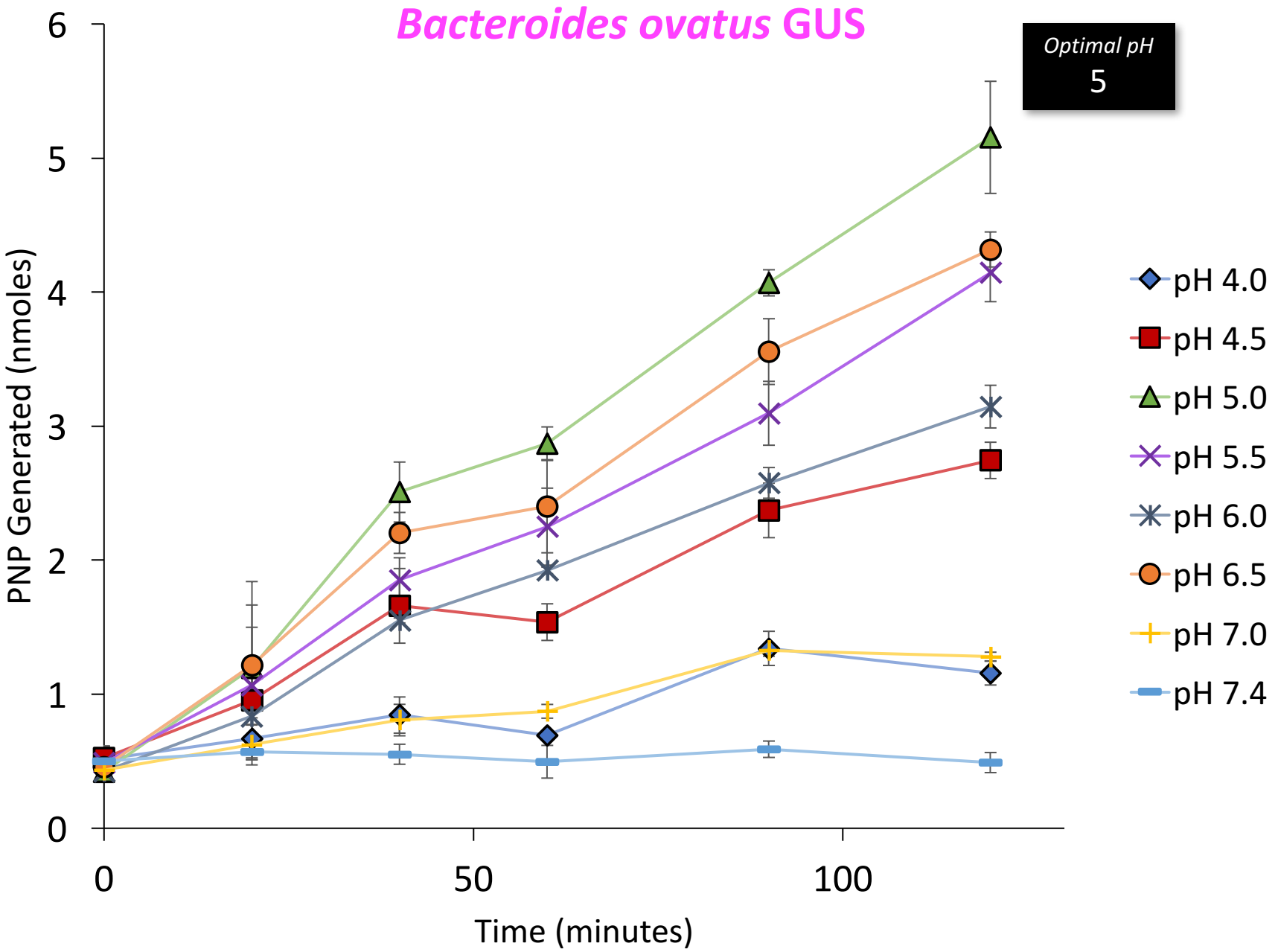


Figure S4H. GUS Panel pH Optima. Related to Figure 5.

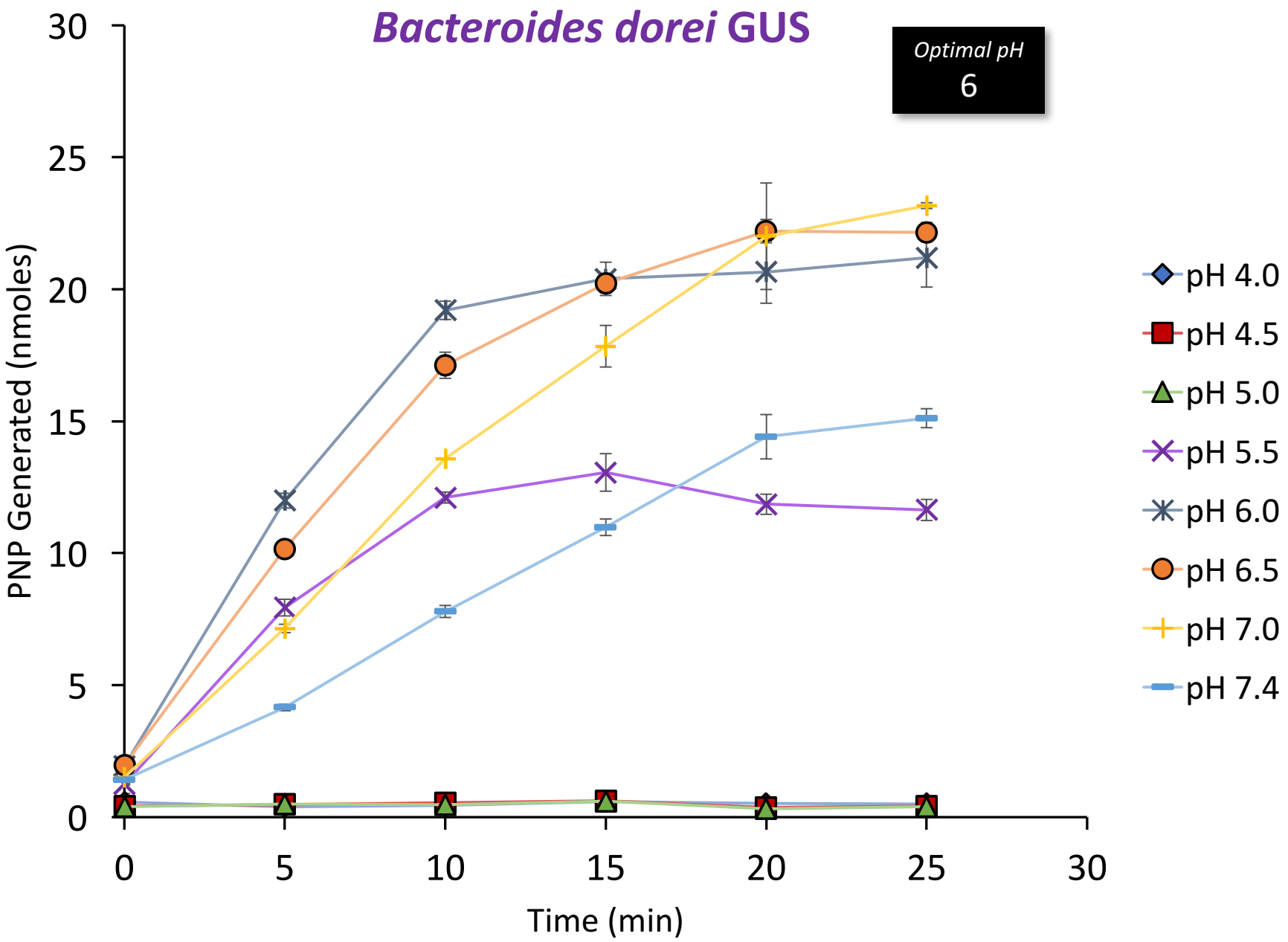
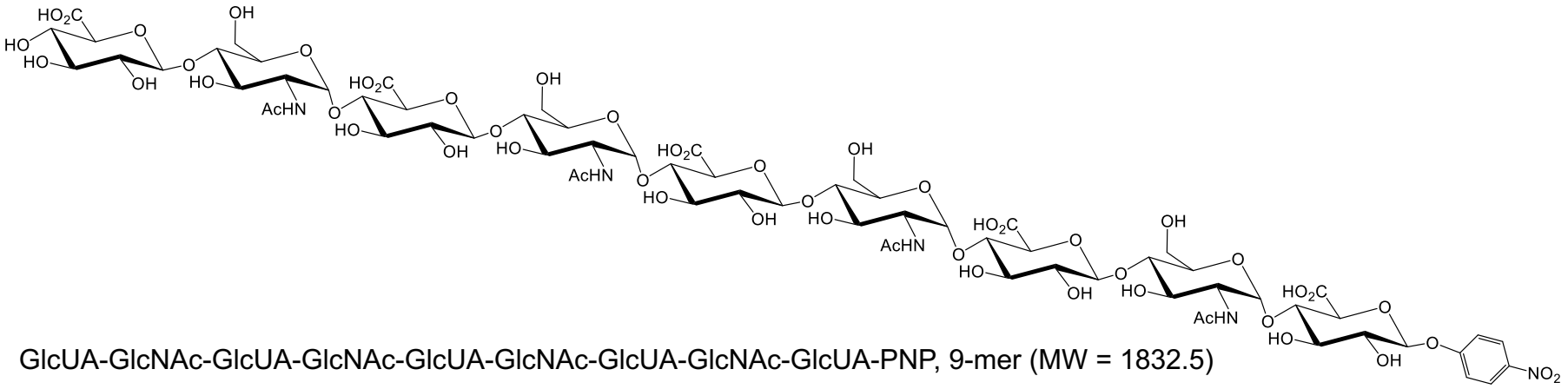


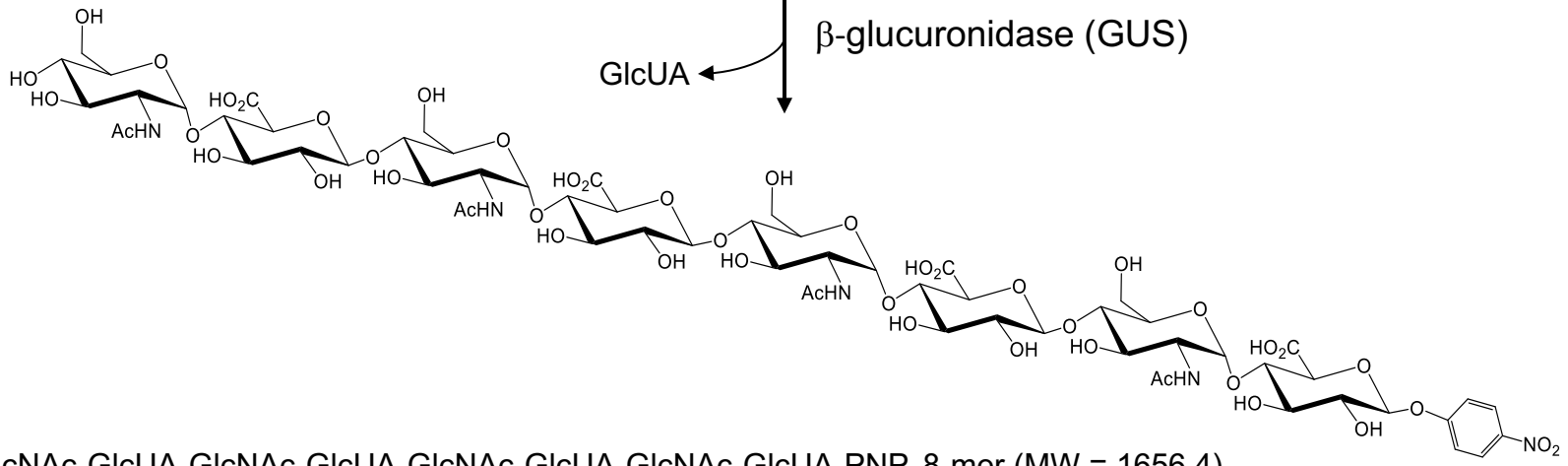
Figure S5A. Heparan Carbohydrate Cleavage by GUS Panel. Related to Figure 5.



GlcUA-GlcNAc-GlcUA-GlcNAc-GlcUA-GlcNAc-GlcUA-GlcNAc-GlcUA-PNP, 9-mer (MW = 1832.5)

β-glucuronidase (GUS)

GlcUA

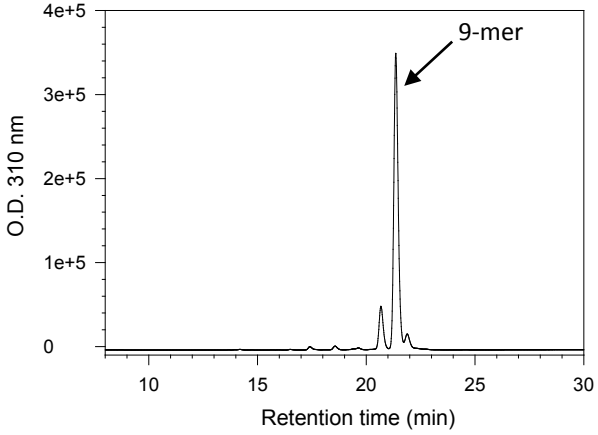


GlcNAc-GlcUA-GlcNAc-GlcUA-GlcNAc-GlcUA-GlcNAc-GlcUA-PNP, 8-mer (MW = 1656.4)

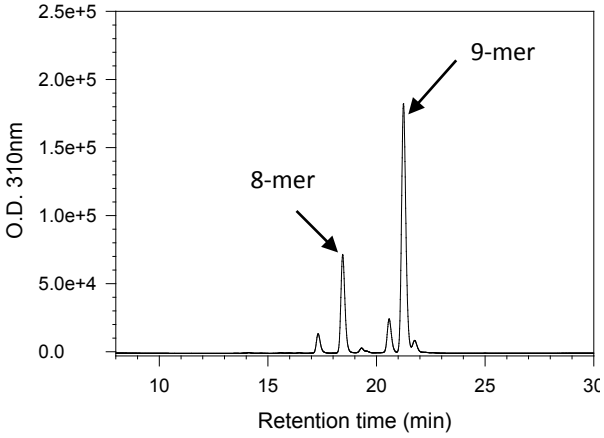
Figure S5B. Heparan Carbohydrate Cleavage by GUS Panel. Related to Figure 5.

Representative HPLC chromatograms and ESI-MS spectra of nonasaccharide digestion

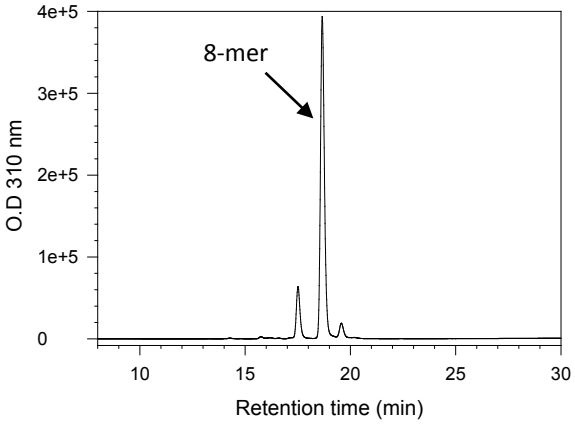
HPLC of uncleaved reaction (*e.g.,* Bo, Ec, Ee, Fp)



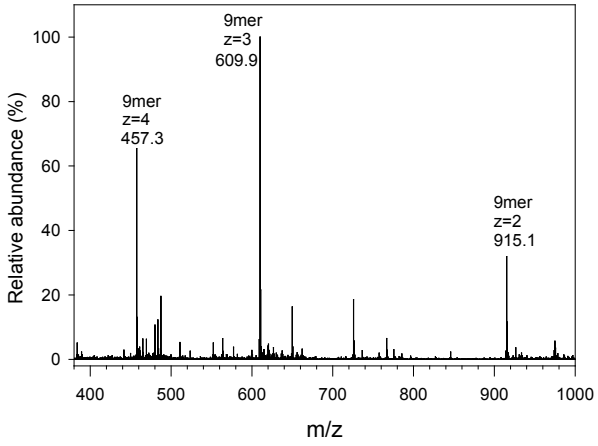
HPLC of partially cleaved reaction (*e.g.,* Bd, Bf, Bu)



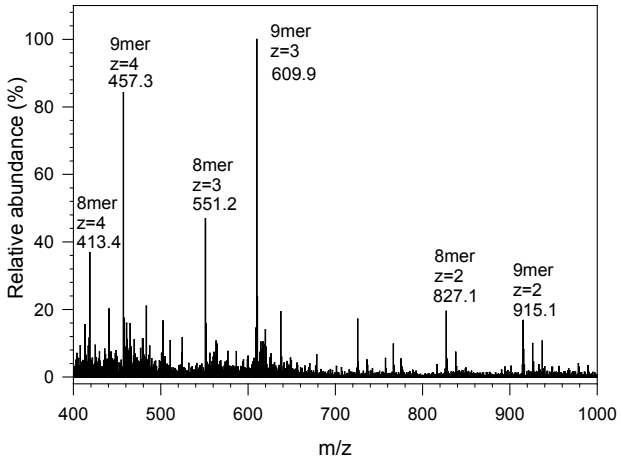
HPLC of fully cleaved reaction (*e.g.,* Pm)



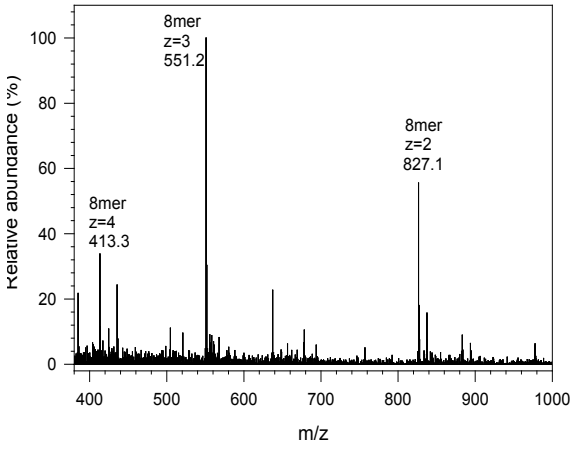
ESI-MS of uncleaved reactions (*e.g.,* Bo,Ec,Ee,Fp)



ESI-MS of partially cleaved reactions (*e.g.,* Bd, Bf, Bu)



ESI-MS of fully cleaved reaction (*e.g.,* Pm)



Supplemental Table 3. Crystallographic Statistics for *B. uniformis* GUS

Resolution (highest shell; Å)	29.6-1.90 (1.93-1.90)
Space group	$P2_12_12_1$
Unit cell dimensions (a, b, c; Å)	74.7, 142.9, 181.7
Total reflections (*F*>0)	1,668,595
Unique reflections	152,573
Multiplicity	10.9 (11.0)
Completeness, % (highest shell)	99.4 (98.7)
Mean I/sigma(I) (highest shell)	21.1 (5.0)
Wilson B-factor ($Å^2$)	22.1
R_{merge} (highest shell)	0.068 (0.448)
R	0.144
R_{free}	0.180
Number of protein, metal, ligand, and water atoms per asymmetric unit	13,655, 4, 36, 1775
rms bonds (Å)	0.011
rms angles (°)	1.28
Ramachandran favored (%)	97.5
Ramachandran outliers (%)	0
Clash score	1.893
Average B-factor ($Å^2$)	30.2
RCSB ID	5UJ6

Table S4. Primer Sequences

Primer Name	Sequence
EeGUS Fwd	TACTTCCAATCCAATGCGATGCTGTACCCTGTCCTGACTCAGTCTCG
EeGUS Rev	TTATCCACTTCCAATGCGCTATTTGGTTTTGTAGCCAAATTCCGGAATGGTGGACCAG
FpGUS Fwd	TACTTCCAATCCAATGCGATGAACCGTAGCCTGCTGTACCCTCGTG
FpGUS Rev	TATCCACTTCCAATGCGCTTTTTTTGCGTTTTTTGAAGTCAACCG
BfGUS Fwd	TACTTCCAATCCAATGCGCGTGAGGATATTCTACTCAACAATAACTGGAATTTCCGTTTT
BfGUS Rev	TTATCCACTTCCAATGCGCTATAGGGGTCTGTCTTCTTTATATTCTACTGTCACTTCGTC
BuGUS Fwd	TACTTCCAATCCAATGCGCAACG TCAAACTCAAACCATCAACGACTCCTGGAAG
BuGUS Rev	TTATCCACTTCCAATGCGCTAATAAATGTTGCGCAGTTTAATGCCATTCAGGAAACACG
PmGUS Fwd	TACTTCCAATCCAATGCGATGAAATACCTGTTCGTCGCTTGCCTGCTGTG
PmGUS Rev	TTATCCACTTCCAATGCGCTATTTCGGGCTTTTCAGTTCCAGAAACGCGGTC
BoGUS Fwd	TACTTCCAATCCAATGCGCAGGAAACTTCTCCGCGTACTATCTTCTCTCTGAAC
BoGUS Rev	TTATCCACTTCCAATGCGCTATTTAATGCAGTACCATTCGCAGGTGTCGGACAG
BdGUS Fwd	TACTTCCAATCCAATGCGAGCGAGATCAGCATCACTGACTCTTGGAAGTATAAG
BdGUS Rev	TTATCCACTTCCAATGCGCTAGCGAATTTTTTTAACTTTCAGGCCGCTAATTACGCCCTG

SUPPLEMENTAL FIGURE, TABLE, AND DATA FILE LEGENDS

Figure S1. Loop Category Variability per Individual. Related to Table 1, Table 2.
For the HMGI3013 GUS analysis, the number of individuals of each loop category is presented, with annotations to indicate averages and percent of individuals with no GUS enzymes of that category, and the averages over all individuals. Panels A-F present this information for the six formal GUS categories (Loop 1, Mini-Loop 1, Loop 2, Mini-Loop 2, Mini-Loop 1,2, No Loop, respectively), while panel G presents this information for the "no content" GUS proteins.

Figure S2. Summary of Loop Variability. Related to Table 2.
Bar graph of the average percent composition of abundance reads for the HMGI3013 dataset within each loop category, with standard deviations of the averages indicated.

Figure S3. GUS Panel Sequence Identities. Related to Figure 5.
Pairwise sequence identities in percent for the GUS panel proteins from *E. coli* (Ec), *Eubacterium eligens* (Ee), *Faecalibacterium prausnitzii* (Fp), *Bacteroides fragilis* (Bf), *B. uniformis* (Bu), *Parabacteroides merdae* (Pm), *B. ovatus* (Bo), and *B. dorei* (Bd). Each enzyme's identifier is colored by loop category: red, Loop 1, green, Mini-Loop 1; blue, Loop 2; orange, Mini-Loop 2; magenta, Mini-Loop 1,2; purple, No Loop.

Figure S4. GUS Panel pH Optima. Related to Figure 5.
Results for amount of PNP produced from PNPG per minute at the eight indicated pH values are presented for each of the eight enzymes in the GUS panel, and colored by loop category as outlined in Figure S3 above. Amount of PNP Generated is shown as the average and standard deviation over three replicates. The pH optimum selected for each GUS is also indicated.

Figure S5. Heparan Carbohydrate Cleavage by GUS Panel. Related to Figure 5.
A. Chemical structures and molecular weights of heparan nonasaccharaide (9-mer) assay substrate and octasaccharide (8-mer) product detected by both HPLC and ESI-MS, where GlcUA is glucuronic acid, GlcNAc is N-acetylglucosamine, and PNP is *p*-nitrophenol. **B**. Representative HPLC chromatograms and ESI-MS spectra for heparan digestion assays with the GUS panel. HPLC results (top) are shown for uncleaved (left), partially cleaved (middle), and fully cleaved (right) reactions. Electrospray ionization mass spectrometry (ESI-MS) spectra are shown for the same types of reactions (left, middle, center). GUS enzymes from the following organisms were examined: Bo, *B. ovatus*, Ec, *E. coli*, Ee, *Eu. eligens*, Fp, *F. prausnitzii*, Bd, *B. dorei*, Bf, *B. fragilis*, Bu, *B. uniformis*, Pm, *P. merdae*.

Table S1. HMGI3013 Characteristics. Related to Table 1, Table 2.
Detailed information is provided for all 3013 GUS proteins identified in the HMGI database, including the sequences of their Loop 1 and Loop 2 regions, protein length in amino acid residues, read abundances, and taxonomy along with BLAST hit characteristics.

Table S2. HMGC279 Characteristics. Related to Table 1, Figure 3.
Detailed information is provided for the 279 GUS proteins identified in the HMGC Clustered Gene Indices database of the HMP, including the sequences of their Loop 1 and Loop 2 regions, protein length in amino acid residues, taxonomy along with BLAST hit characteristics, and signal peptide search results.

Table S3. BuGUS Crystallographic Analysis. Related to Figure 5.
X-ray data collection, structure determination, and structure refinement information are provided for the crystal structure of the Loop 2 GUS from *B. uniformis* (BuGUS).

Table S4. Primer Sequences. Related to Figure 5.
Forward and reverse primers employed in this study.

Data File S1. HMGI3013 Sequences. Related to Figure 1.
FASTA sequences for the 3013 total GUS proteins identified in the HMGI Gene Indices database, part of the Stool Sample Catalog of the Human Microbiome Project (HMP). Entries are labeled by their HMP identifier.

Data File S2. HMGC279 Sequences. Related to Figure 1, Figure 3, Table 1.
FASTA sequences for the 279 unique GUS proteins identified in the HMGC Clustered Gene Indices database, part of the Stool Sample Catalog of the Human Microbiome Project (HMP). Entries are labeled by their HMP identifier, along with the GUS category it belongs to (L1, mL1, L2, mL2, mL1,2, NL, nc), and taxonomy when identified.

Data File S3. Representative GUS Loop Alignment. Related to Figure 2.
Multiple sequence alignment of the amino acid sequence of twelve representative GUS enzymes, with residue numbers indicated at right, and loop regions colored as follows: red, Loop 1; green, Mini-Loop 1; blue, Loop 2; orange, Mini-Loop 2; magenta, Mini-Loop 1, 2; purpe, No Loop. The twelve GUS enzymes are from *Homo sapiens* (HsGUS), *Faecalibacterium prausnitzii* (FpGUS), *E. coli* (EcGUS), *Streptococcus agalactiae* (SaGUS), *Clostridium perfringens* (CpGUS), *Eubacterium eligens* (EeGUS), *Parabacteroides merdae* (PmGUS), a Firmicutes GUS described from human fecal metagenomic sequencing (H11G11-BG), *Bacteroides fragilis* (BfGUS), *B. ovatus* (BoGUS), *B. uniformis* (BuGUS), and *B. dorei* (BdGUS). Identical amino acid positions are indicated with an asterisk (*), while largely conserved and conserved amino acid positions are indicated with a colon (:) or period (.), respectively.